

Data Sheet

ENERCON SCADA PDI-OPC

ENERCON EP1, EP2, EP3, EP4 Wind Energy Converters

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D0137978	Technische Beschreibung ENERCON SCADA PDI-OPC (Technical description of ENERCON SCADA PDI-OPC)
D0179325	Technische Beschreibung ENERCON SCADA Datei- und Feldnamenbeschreibung (Technical description of ENERCON SCADA file and field name description)

Table of contents

1	Introduction.....	6
2	Historical data	7
3	Nodes and items for data querying.....	9
3.1	The Loc node	9
3.2	Loc\Wec node	10
3.2.1	Loc\Wec\Plant<N> node	12
3.2.2	Loc\Wec\Plant<N>\Status node	14
3.2.3	Loc\Wec\Plant<N>\Log node	16
3.2.3.1	Loc\Wec\Plant<N>\Log\Wec15 node (optional)	17
3.2.3.2	Loc\Wec\Plant<N>\Log\Wec15\<Period> node (optional)	18
3.2.3.3	Loc\Wec\Plant<N>\Log\T<XX> node	19
3.2.3.4	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node	20
3.2.3.5	Loc\Wec\Plant<N>\Log\Wecstd node	33
3.2.3.6	Loc\Wec\Plant<N>\Log\Wecstd\<Period> node.....	34
3.2.3.7	The Loc\Wec\Plant<N>\Log\Weceolog node (optional)	36
3.2.3.8	The Loc\Wec\Plant<N>\Log\Weceolog\<Period> node (optional)	36
3.2.3.9	The Loc\Wec\Plant<N>\Log\Wecfos4x node (optional)	39
3.2.3.10	The Loc\Wec\Plant<N>\Log\Wecfos4x\<Period> node (optional).....	39
3.2.3.11	The Loc\Wec\Plant<N>\Log\Wecwoef node (optional)	40
3.2.3.12	The Loc\Wec\Plant<N>\Log\Wecwoef\<Period> node (optional).....	40
3.2.4	Loc\Wec\Plant<N>\Ava node	41
3.2.4.1	Loc\Wec\Plant<N>\Ava\<Period> node	43
3.2.5	Loc\Wec\Plant<N>\Para node.....	43
3.3	Loc\Sens node	44
3.3.1	Loc\Sens\Plant<N> node	44
3.3.2	Loc\Sens\Plant<N>\Log\Mspgerf1\<Period> node.....	45
3.4	Loc\Conv node.....	46
3.4.1	Solar converter	46
3.4.1.1	Loc\Conv\Plant<N>\Log\Solar1\<Period> node	47
3.4.2	Feeding and charging unit.....	49
3.4.2.1	Loc\Conv\Plant<N>\Log\Bcs1\<Period> node	50
3.4.2.2	Loc\Conv\Plant<N>\Log\Bcs2\<Period> node	51
3.5	Loc\Ctrl node.....	56
3.5.1	RTU	56
3.5.1.1	Loc\Ctrl\Plant<N>\Log\Rtu1\<Period> node.....	58
3.5.1.2	Loc\Ctrl\Plant<N>\Log\Rtu15\<Period> node (optional).....	60
3.5.1.3	Loc\Ctrl\Plant<N>\Log\Rtuphase1\<Period> node	61
3.5.1.4	Loc\Ctrl\Plant<N>\Para node of the RTU	62

3.5.2	FCU	63
3.5.2.1	Loc\Ctrl\Plant<N>\Log\Fcuctr1\<Period> node	64
3.5.3	FCU E2.....	66
3.5.3.1	Loc\Ctrl\Plant<N>\Log\Fcu2ctr1\<Period> node	67
3.5.4	GDA.....	71
3.5.4.1	Loc\Ctrl\Plant<N>\Log\Farmctr1\<Period> node	73
3.5.4.2	Loc\Ctrl\Plant<N>\Log\Farmctr15<Period> node (optional)	75
3.5.5	SCU.....	76
3.5.5.1	Loc\Ctrl\Plant<N>\Log\Substat1\<Period> node	77
3.5.6	Transmission substation 2.....	78
3.5.6.1	Loc\Ctrl\Plant<N>\Log\Substat2\<Period> node	80
3.5.6.2	Loc\Ctrl\Plant<N>\Log\SubTemp2\<Period> node.....	82
3.5.6.3	Loc\Ctrl\Plant<N>\Log\Subphase2\<Period> node	83
3.6	Loc\Meteo node	84
3.6.1	Loc\Meteo\Plant<N> node.....	84
3.6.2	Loc\Meteo\Plant<N>\Log\Wind1\<Period> node.....	85
3.6.3	Loc\Meteo\Plant<Index>\Log\Weather2\<Period> node	86
3.7	Loc\Actor node	87
3.7.1	PDI	87
3.7.1.1	Loc\Actor\Plant<N>\Para node of the PDI	88
3.8	The Doc (documentation) node.....	88
3.9	The SCADA node	89
4	Nodes and items for closed-loop and open-loop control	90
4.1	Session items for access for control purposes.....	90
4.1.1	Session items for access to individual WECs for control purposes.....	92
4.1.2	Session items for wind farm start/stop of all wind energy converters in a wind farm	96
4.1.3	Session items for resetting the control system of individual wind energy converters	97
4.2	Items for wind farm closed-loop and open-loop control (ENERCON SCADA Server)	98
4.2.1	Weekly table control	98
4.2.2	Farm Control 2	100
4.2.3	Apparent power control 2	103
4.2.4	Open Loop Control (OIctr1)	104
4.2.5	Manual control 1 (ManCtr1).....	108
4.2.6	Power Consumption Management.....	111
4.3	Items for closed-loop and open-loop RTU/FCU/FCU E2 control systems	115
	List of tables.....	119
	Glossary of terms	122

1 Introduction

The ENERCON SCADA PDI-OPC is an OPC XML-DA server in accordance with the OPC Foundation specifications V1.01. Using the ENERCON SCADA PDI-OPC, wind farm data can be analysed online and control values can be transmitted to the wind farm without having to use the ENERCON SCADA Remote software.

This document describes the data tree and the OPC items of the ENERCON SCADA PDI-OPC. Refer to document D0179325 "Technische Beschreibung ENERCON SCADA Datei- und Feldnamenbeschreibung" (Technical description of ENERCON SCADA file and field name description) for a description of the abbreviations used in the ENERCON SCADA PDI-OPC.

This document describes both the standard scope of supply and the optional components. Which of the information and data contained in this document can be called up and/or used by each customer depends on the systems installed in the farm.

Unless otherwise specified, the WEC-related information and data contained in this document are applicable and valid for the following ENERCON wind energy converters:

E-44, E-48, E-53, E-70 E4, E-82 E2, E-82 E4, E-92, E-103 EP2, E-101, E-115, E-115 E2, E-126 EP3, E-115 EP3 E3, E-138 EP3, E-138 EP3 E2, E-126 EP4 and E-141 EP4.



Due to potential communication problems or disruptions, it is not possible to guarantee fault-free, complete data transmission to the WECs in every case. The start/stop function is subject to this restriction.

2 Historical data

The following historical data is available via OPC:

Standard data

Fcuctr1, Fcu2ctr1, Fcu2ctr2, Farmctr1, Mspgerf1, Rtu1, Rtuphase1, Solar1, Substat1, Substat2, SubTemp2, Subphase2, Waecstd1, Wecstd, Wind1 and Weather2.

Tab. 1: Standard data

Data type	Number of data sets	Sampling rate	Timeframe
Raw data (Raw)	720	60 s	12 hours
Report data (Rep)	288	600 s	48 hours
Daily data (Day)	61	1 day	60 days + current day
Weekly data (Week)	55	1 week	54 weeks + current week
Monthly data (Month)	37	1 month	36 months + current month
Annual data (Year)	11	1 year	10 years + current year

Tab. 2: 15-minute data (Wec15, Rtu15, Farmctr15) (optional)

Data type	Number of data sets	Sampling rate	Timeframe
Raw data (Raw)	720	60 s	12 hours
Report data (Rep)	288	900 s	72 hours

Tab. 3: Availability data (Ava)

Data type	Number of data records	Sampling rate	Time frame
Daily data	91	1 day	90 days + current day
Weekly data	25	1 week	24 weeks + current week
Monthly data	13	1 month	12 months + current month
Yearly data	11	1 year	10 years + current year

Tab. 4: Temperature data (TXX)

Data type	Number of data sets	Sampling rate	Timeframe
Raw data (Raw)	60	60 s	1 hour
Report data (Rep.)	288	600 s	48 hours
Daily data (Day)	61	1 day	60 days + current day

Status data

PDI-OPC retains the 200 most current status messages.



If the ENERCON SCADA Server cannot retrieve any data from the plants and devices, e.g. due to a communication fault, no data will be available for this time period.

3 Nodes and items for data querying

In OPC, signs (+ and -) are used in accordance with IEC 61400-21.

3.1 The Loc node

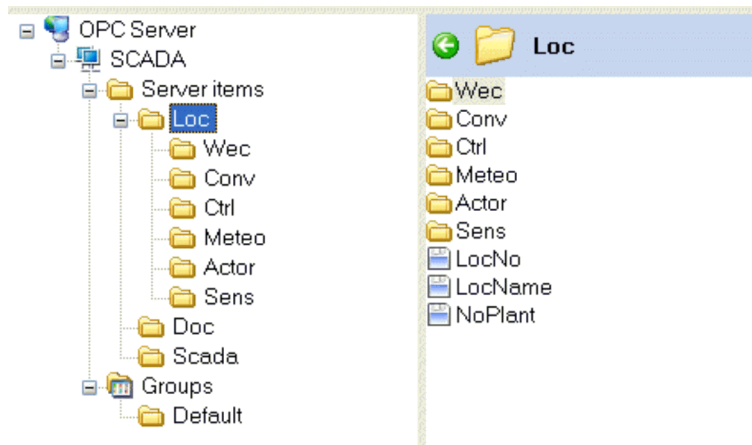


Fig. 1: Loc node

This node contains the data for all the systems in one wind farm.

Tab. 5: Loc node

Node/item	Description
Wec	Data from the wind energy converters
Conv	Data from the converter systems (e.g. solar converter)
Ctrl	Data from wind-farm control systems (e.g. RTU, FCU E2)
Meteo	Data from the ENERCON SCADA METEO weather-recording system
Actor	Data from actors (e.g. PDI)
Sens	Data from the sensor systems (e.g. medium-voltage data acquisition)
LocNo	Wind farm number assigned by ENERCON
LocName	Wind farm name assigned by the customer
NoPlant	Total number of all plants in the wind farm

3.2 LocWec node

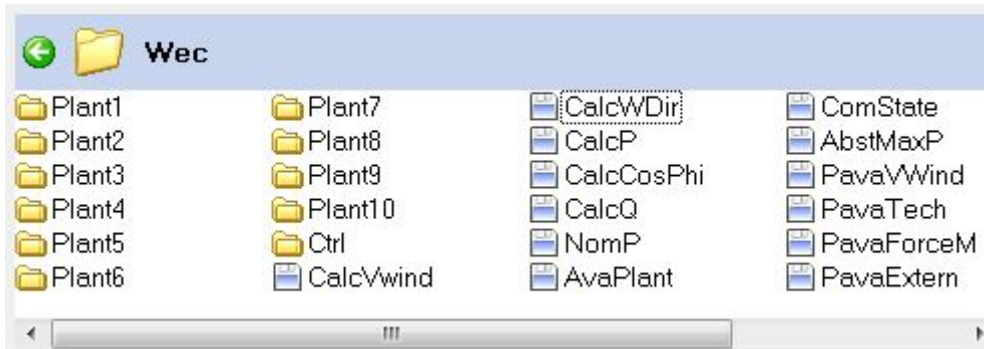


Fig. 2: LocWec node

This node contains the data of plant group *wind energy converters*.

Tab. 6: LocWec node

Node/item	Description
Plant<N>	Data from wind energy converter <N>
Ctrl	Wind-farm control (start/stop)
CalcVwind	Mean wind speed across all wind energy converters in m/s
CalcWDir	Mean wind direction across all wind energy converters in ° (0° = North)
CalcP	Total power across all wind energy converters in kW
CalcCosPhi	Mean Cos Phi power factor across all wind energy converters
CalcQ	Total reactive power across all wind energy converters in kVAR
NomP	Total power of wind energy converters including the reserve power for compensating inherent losses in kW
AvaPlant	Available wind energy converters in % (wind energy converters that are able to feed energy into the grid)
ComState	Available plant communication in % Plant communication will be considered inactive if three successive data-query cycles have failed.
AbstMaxP	Total theoretically available active power of the wind energy converters in kW. In wind energy converters with older ¹⁾ control software, this value corresponds to a value calculated by the ENERCON SCADA Server. In wind energy converters with newer ²⁾ control software, this value corresponds to the sum of the smallest available power calculated for each wind energy converter (PavaVWind, PavaTech or PavaForceM). In mixed wind farms, i.e. farms with a combination of older and newer control software, this value corresponds to a value calculated by the ENERCON SCADA Server.
PavaVWind ²	Total theoretically available active power of wind energy converters in the wind in kW. This value corresponds to the power the wind energy converters could theoretically feed into the grid in the current wind conditions, assuming the technical availability of all internal and external equipment and the absence of power output limitations through external setpoints.

Node/item	Description
PavaTech ²	Total technically available active power of wind energy converters in kW. This value corresponds to the power the wind energy converters can feed into the grid in spite of technical problems (generator overtemperature; defective converter, etc.). If there is no power limitation for technical reasons, this value equals the theoretically available active power PavaVWind.
PavaForceM ²	Total maximum available active power of the wind energy converters limited by force majeure in kW. This value corresponds to the power the wind energy converters can feed into the grid in case of power limitations due to force majeure (storm control, grid overvoltage, etc.). If there is no power limitation due to force majeure, this value equals the theoretically available active power PavaVWind.
PavaExtern ²	Total available active power of the wind energy converters limited by external setpoints in kW. This value corresponds to the power the wind energy converters can feed into the grid in case of power limitations due to external setpoints. If there is no power limitation due to external setpoints, this value equals the theoretically available active power PavaVWind.

¹ Wind energy converters with control software: CS12, CS16, CS17, CS20, CS30, CS31a, CS32, CS40, CS44a, CS44b, CS58a, CS58b, CS66, CS70a and CS70b

² Wind energy converters with control software: CS48, CS82, CS101, CS126, EP3-CS-02 and EP4-CS-01

3.2.1 Loc\Wec\Plant<N> node

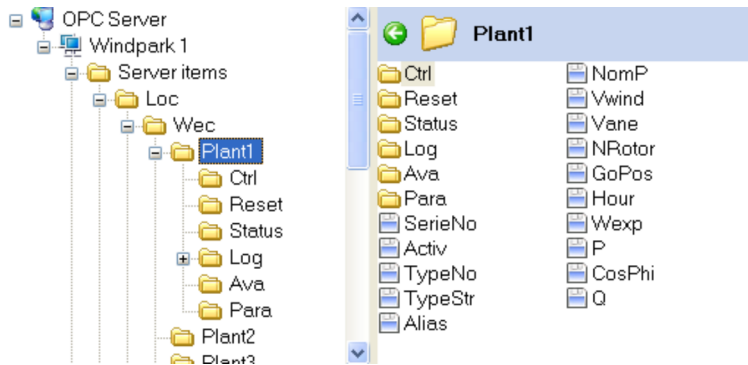


Fig. 3: Loc\Wec\Plant<N> node

This node contains all data of an individual wind energy converter.

Tab. 7: The Loc\Wec\Plant<N> node

Node/item	Description
Ctrl	Control system of this wind energy converter (start/stop)
Reset	Reset of the control system
Status	Status data for this wind energy converter
Log	Report data for this wind energy converter
Ava	Availability data for this wind energy converter
Para	Parameters for this wind energy converter
SerieNo	ENERCON serial number
Activ	Plant is active in the ENERCON SCADA system. In passive mode, the plant is registered with the system, although none of the processes take the plant into account. This mode is used for commissioning and during lengthy service operations.
TypeNo	ENERCON control system type number
TypStr	ENERCON control system type string
Alias	ID that can be selected by the customer (10 characters)
NomP	Nominal power of the wind energy converter, including the reserve power for compensating inherent losses in kW
Vwind	Wind speed at wind energy converter in m/s
Vane	Wind vane of wind energy converter in ° (difference between wind direction and nacelle position)
NRotor	Rotor speed of wind energy converter in rpm (online value)
GoPos	Nacelle position of the wind energy converter in ° (online value)
Hour	Operating hours of the wind energy converter in h (online value)
Wexp	Energy meter of the wind energy converter in kWh (online value)
P	Power of the wind energy converter in kW (online value)
CosPhi	Power factor Cos Phi at wind energy converter (online value)
Q	Reactive power of wind energy converter in kVAR (online value)

Node/item	Description
U-L1	Low voltage U12 in V (online value)
U-L2	Low voltage U23 in V (online value)
U-L3	Low voltage U31 in V (online value)
I-L1	Power feed L1 in A (online value)
I-L2	Power feed L2 in A (online value)
I-L3	Power feed L3 in A (online value)
Freq	Grid frequency in Hz (online value)
PavaVWind ¹	Theoretically available active power of the wind energy converter in the wind in kW
PavaTech ¹	Technically available active power of the wind energy converter in kW
PavaForceM ¹	Maximum available active power of the wind energy converter limited by force majeure in kW
PavaExtern ¹	Available active power of the wind energy converter limited by external setpoints in kW

¹ Only available for wind energy converters with CS48, CS82, CS101, CS126, EP3-CS-02 and EP4-CS-01 control software. For a detailed description of the PavaVWind, PavaTech, PavaForceM and PavaExtern nodes, see ch. 3.2, p. 10.



The number <N> of each plant is assigned during commissioning and remains the same for that plant's entire service life. The numbering cannot be changed because of its reference to data that is key to the ENERCON PartnerKonzept and warranties associated with it. Customers can use (and change) the selected alias to individually identify the plant.

3.2.2 Loc\Wec\Plant<N>\Status node

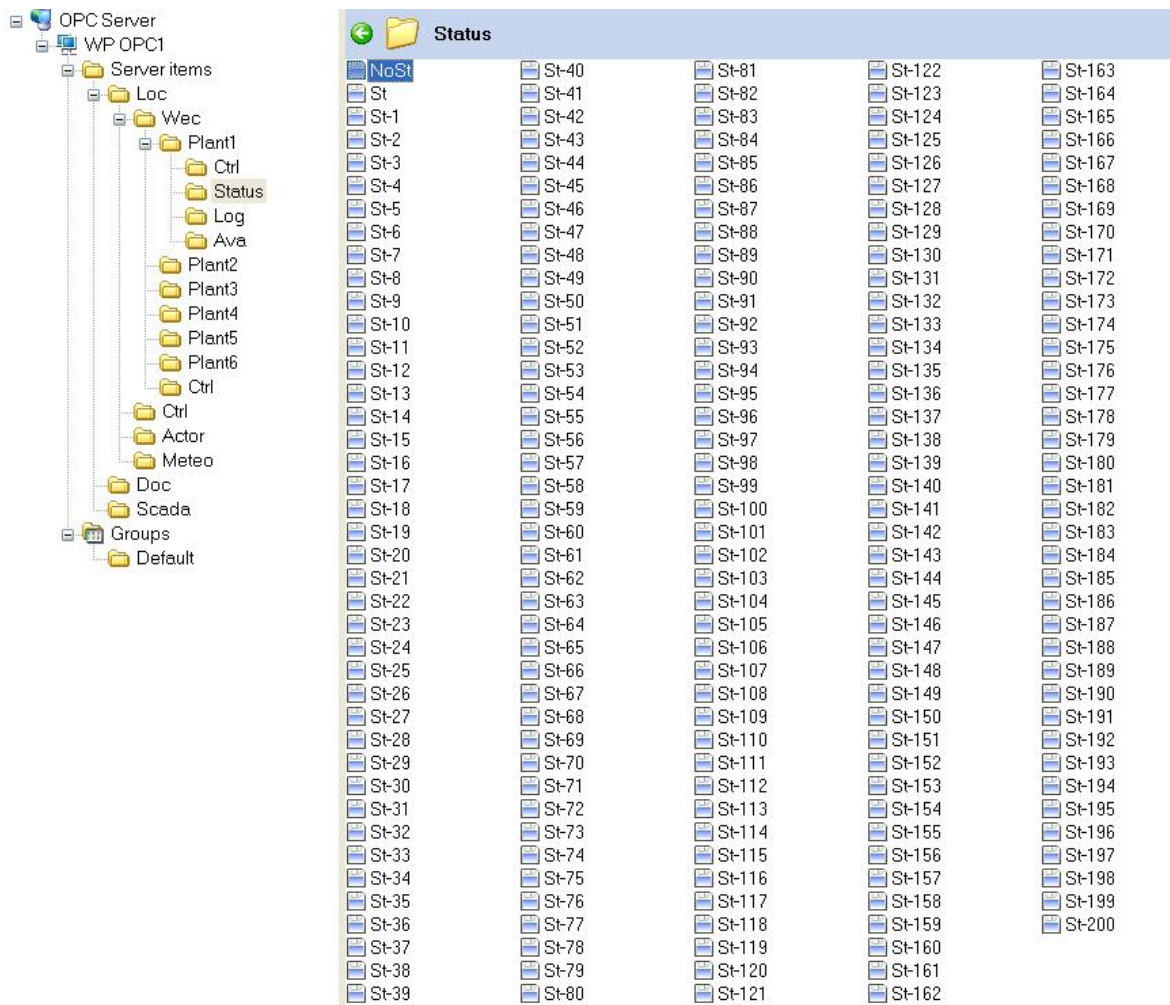


Fig. 4: Loc\Wec\Plant<N>\Status node

The *Status* node contains current and historical statuses of wind energy converter <N>. This data belongs to the report data group.

This node exists for all plant groups. The following is an explanation of the node for one wind energy converter <N> only.

The data records *ST-<X>* are arrays and indicate the preceding statuses. Item *St* indicates the current status. The maximum number of historical statuses stored in the ENERCON SCADA System is specified by Item *NoSt*; this value is limited to a maximum of 200 status messages.

Tab. 8: Node Loc\Wec\Plant<N>\Status – Items

Item	Description
NoSt	Number of available historical statuses (maximum: 200)
St	Current status
St-1	Historical status
St-2 to 200	Historical statuses in chronological order

The elements of the array (ID 2) for items *St* and *St-<N>* have the following meanings:

Tab. 9: Node *Loc\Wec\Plant<N>\Status* – Elements of the array

Index	Description
0	Main status
1	Substatus
2	Fault message flag
3	Information
4	Additional information
5	Warning message flag
6	Service flag

A list of status messages for individual WEC types is available from ENERCON Service on request.

3.2.3 Loc\Wec\Plant<N>\Log node

The name of the *Log* node is derived from an abbreviation for 'data logger'. This node contains report data for plant <N> and its history. The node contains nodes for the wind energy converter's standard data (*Wecstd*) as well as temperatures. If the temperature data is recorded by more than one data logger, the data loggers are shown in sequential order (e.g. *T82a1* and *T82a2* for plants with the CS82 control software).

The node contains all data that is also stored on the ENERCON SCADA Server as 10-minute, hourly, daily, weekly, monthly, and annual report data.

Optionally, the *Log* node can also contain selected 15-minute data (e.g. *Wec15* node).

Tab. 10: The Loc\Wec\Plant<N>\Log node

Node	Description
Wec15	15-minute data of the wind energy converter (optional)
TXX	Temperatures of the wind energy converter T40 = Wind energy converters with CS40 control software T44a = Wind energy converters with CS44a control software T44b = Wind energy converters with CS44b control software T48 = Wind energy converters with CS48 control software T66 = Wind energy converters with CS66 control software T70a = Wind energy converters with CS70a control software T70b = Wind energy converters with CS70b control software T82a1 and T82a2 = Wind energy converters with CS82 control software T101a1 and T101a2 = Wind energy converters with CS101 control software T126a1, T126a2 and T126a3 = Wind energy converters with CS126 control software Tep3c021 and Tep3c022 = Wind energy converters with EP3-CS-02 control software Tep4c011 and Tep4c012 = Wind energy converters with EP4-CS-01 control software
Wecstd	Standard data of wind energy converter
Weceolog	eologix ice detection system data
Wecfos4x	fos4X ice detection system data
Wecwoef	Wölfel ice detection system data

3.2.3.1 Loc\Wec\Plant<N>\Log\Wec15 node (optional)

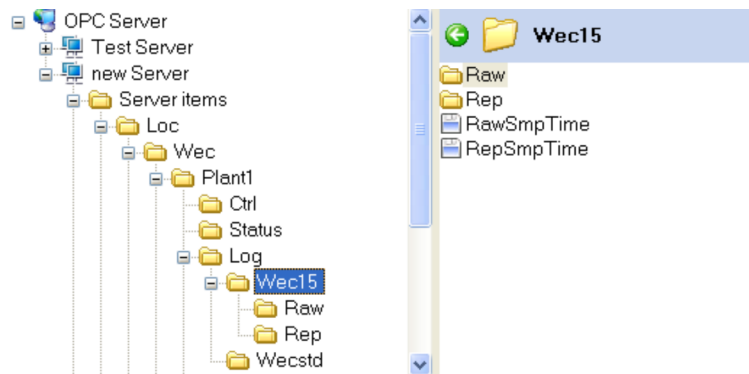


Fig. 5: Loc\Wec\Plant<N>\Log\Wec15 node (optional)

The optional *Loc\Wec\Plant<N>\Log\Wec15* node displays 15-minute data.

Tab. 11: Loc\Wec\Plant<N>\Log\Wec15 node (optional)

Node/Item	Description
Raw	Raw data (precalculated data)
Rep	Report data
RawSmpTime	Raw data sampling rate in s (default: 60 s)
RepSmpTime	Report data sampling rate in s (default: 900 s)

3.2.3.2 Loc\Wec\Plant<N>\Log\Wec15\<Period> node (optional)

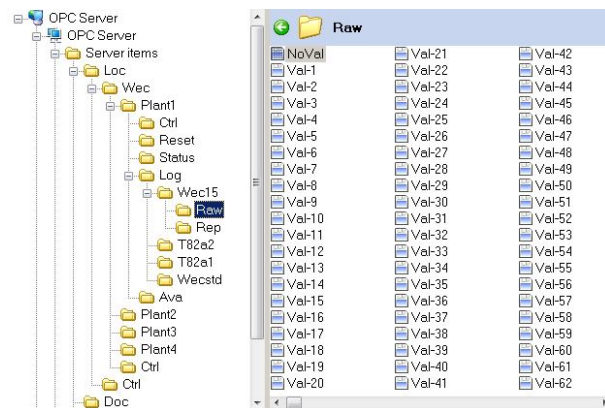


Fig. 6: Loc\Wec\Plant<N>\Log\Wec15\Raw node (optional)

The *NoVal* item shows the number of written data records.

The data records *Val- X* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 12: Loc\Wec\Plant<N>\Log\Wec15\<Period> node (optional)

Description	Raw	Rep
Number of sampling values	0	0
Average wind speed in m/s	1	1
Average active power in kW	2	2
Average nacelle position including cable twisting in °	3	3
Operating hours in h	4	4
Energy production in kWh	5	5
Average reactive power in kvar	6	6

3.2.3.3 Loc\Wec\Plant<N>\Log\T<XX> node

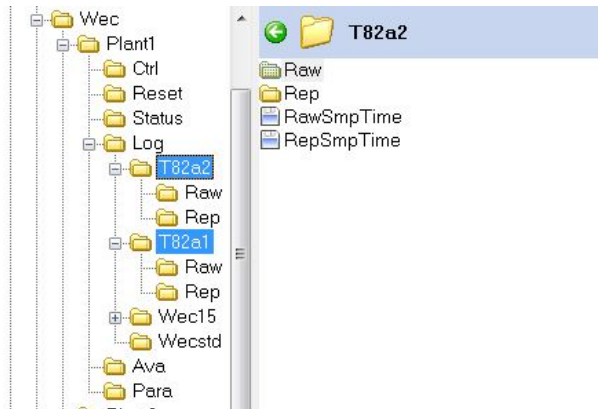


Fig. 7: Nodes *Loc\Wec\Plant<N>\Log\T82a1* and *T82a2*

This *T<XX>* node contains temperature data for the wind energy converter. The name and the contents of the node depend on the control software used (e.g. *T82a1* and *T82a2* for WECs with CS82 control software).

Tab. 13: *Loc\Wec\Plant<N>\Log\T<XX>* node

Node/item	Description
Raw	Raw data (pre-calculated data)
Rep	Report data
Day	Daily reports
RawSmpTime	Raw-data sampling rate in s (default: 60 s)
RepSmpTime	Report-data sampling rate in s (default: 600 s)

3.2.3.4 Loc\Wec\Plant<N>\Log\T<XX>\<Period> node

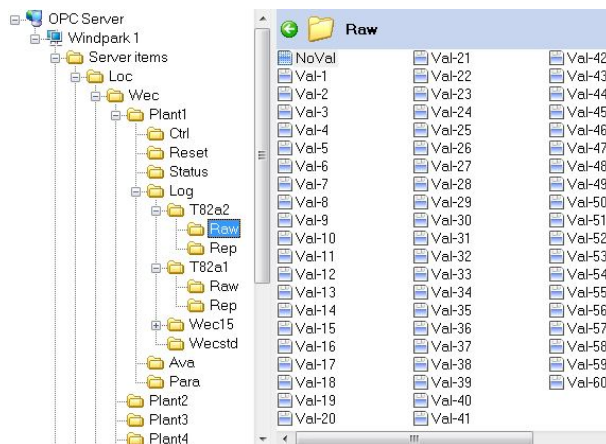


Fig. 8: Loc\Wec\Plant<N>\Log\T82a2\Raw node

The *NoVal* item shows the number of written data records.

The data records *Val- X* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

The temperature data for all WEC control system types are shown below.

Tab. 14: Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS40 (T40)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Stator temperature in °C	1	1	1
Rotor temperature in °C	2	2	2
Nacelle ambient temperature in °C	3	3	3
Front bearing temperature in °C	4	4	4
Rear bearing temperature in °C	5	5	5
Tower temperature in °C	6	6	6
Control cabinet temperature in °C	7	7	7
Rectifier cabinet temperature 1 in °C	8	8	8
Rectifier cabinet temperature 2 in °C	9	9	9
Blade A temperature in °C	10	10	10
Blade B temperature in °C	11	11	11
Blade C temperature in °C	12	12	12

Tab. 15: Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS44a (T44a)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Stator temperature in °C	1	1	1
Rotor temperature in °C	2	2	2

Description	Raw	Rep	Day
Nacelle ambient temperature in °C	3	3	3
Nacelle temperature in °C	4	4	4
Front bearing temperature in °C	5	5	5
Rear bearing temperature in °C	6	6	6
Blade A temperature in °C	7	7	7
Blade B temperature in °C	8	8	8
Blade C temperature in °C	9	9	9
Nacelle control cabinet temperature in °C	10	10	10
Control cabinet temperature in °C	11	11	11
Tower temperature in °C	12	12	12
Transformer temperature in °C	13	13	13
Ambient temperature in °C	14	14	14
Yaw inverter cabinet temperature in °C	15	15	15

Tab. 16: Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS44b (T44b)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Stator temperature in °C	1	1	1
Rotor temperature in °C	2	2	2
Nacelle ambient temperature in °C	3	3	3
Nacelle temperature in °C	4	4	4
Front bearing temperature in °C	5	5	5
Rear bearing temperature in °C	6	6	6
Blade A temperature in °C	7	7	7
Blade B temperature in °C	8	8	8
Blade C temperature in °C	9	9	9
Nacelle control cabinet temperature in °C	10	10	10
Control cabinet temperature in °C	11	11	11
Tower temperature in °C	12	12	12
Transformer temperature in °C	13	13	13
Ambient temperature in °C	14	14	14
Yaw inverter cabinet temperature in °C	15	15	15

Tab. 17: Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS48 (T48a)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Spinner temperature in °C	1	1	1

Description	Raw	Rep	Day
Front bearing temperature in °C	2	2	2
Rear bearing temperature in °C	3	3	3
Blade A pitching heat sink temperature in °C	4	4	4
Blade B pitching heat sink temperature in °C	5	5	5
Blade C pitching heat sink temperature in °C	6	6	6
Blade A pitch control box temperature in °C	7	7	7
Blade B pitch control box temperature in °C	8	8	8
Blade C pitch control box temperature in °C	9	9	9
Blade A temperature in °C	10	10	10
Blade B temperature in °C	11	11	11
Blade C temperature in °C	12	12	12
Rotor temperature 1 in °C	13	13	13
Rotor temperature 2 in °C	14	14	14
Stator temperature 1 in °C	15	15	15
Stator temperature 2 in °C	16	16	16
Nacelle ambient temperature in °C	17	17	17
Nacelle temperature in °C	18	18	18
Nacelle control cabinet temperature in °C	19	19	19
Rectifier heat sink temperature 1 in °C	20	20	20
Rectifier heat sink temperature 2 in °C	21	21	21
Rectifier cabinet temperature in °C	22	22	22
Excitation controller box heat sink temperature in °C	23	23	23
Ambient temperature in °C	24	24	24
Tower temperature in °C	25	25	25
Control cabinet temperature in °C	26	26	26
Transformer temperature in °C	27	27	27
Sys. 1 power cabinet temperature 1 in °C	28	28	28
Sys. 1 power cabinet temperature 2 in °C	29	29	29
Sys. 1 power cabinet temperature 3 in °C	30	30	30

Tab. 18: Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS66 (T66)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Stator temperature in °C	1	1	1
Rotor temperature in °C	2	2	2
Nacelle ambient temperature in °C	3	3	3

Description	Raw	Rep	Day
Nacelle temperature in °C	4	4	4
Spinner temperature in °C	5	5	5
Front bearing temperature in °C	6	6	6
Rear bearing temperature in °C	7	7	7
Blade A temperature in °C	8	8	8
Blade B temperature in °C	9	9	9
Blade C temperature in °C	10	10	10
Nacelle control cabinet temperature in °C	11	11	11
Control cabinet temperature in °C	12	12	12
Tower temperature in °C	13	13	13
Transformer temperature in °C	14	14	14
Sys. power cabinet temperature 1 in °C	15	15	15
Sys. power cabinet temperature 2 in °C	16	16	16
Sys. power cabinet temperature 3 in °C	17	17	17
Sys. power cabinet temperature 4 in °C	18	18	18
Sys. power cabinet temperature 5 in °C	19	19	19
Sys. power cabinet temperature 6 in °C	20	20	20
Ambient temperature in °C	21	21	21

Tab. 19: Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS70a (T70a)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Stator temperature in °C	1	1	1
Rotor temperature in °C	2	2	2
Nacelle ambient temperature in °C	3	3	3
Nacelle temperature in °C	4	4	4
Spinner temperature in °C	5	5	5
Front bearing temperature in °C	6	6	6
Rear bearing temperature in °C	7	7	7
Blade A temperature in °C	8	8	8
Blade B temperature in °C	9	9	9
Blade C temperature in °C	10	10	10
Nacelle control cabinet temperature in °C	11	11	11
Control cabinet temperature in °C	12	12	12
Tower temperature in °C	13	13	13
Transformer temperature in °C	14	14	14

Description	Raw	Rep	Day
Ambient temperature in °C	15	15	15
Yaw inverter cabinet temperature in °C	16	16	16

Tab. 20: *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for CS70b (T70b)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Stator temperature in °C	1	1	1
Rotor temperature in °C	2	2	2
Nacelle ambient temperature in °C	3	3	3
Nacelle temperature in °C	4	4	4
Spinner temperature in °C	5	5	5
Front bearing temperature in °C	6	6	6
Rear bearing temperature in °C	7	7	7
Blade A temperature in °C	8	8	8
Blade B temperature in °C	9	9	9
Blade C temperature in °C	10	10	10
Nacelle control cabinet temperature in °C	11	11	11
Control cabinet temperature in °C	12	12	12
Tower temperature in °C	13	13	13
Transformer temperature in °C	14	14	14
Ambient temperature in °C	15	15	15
Yaw inverter cabinet temperature in °C	16	16	16

Tab. 21: *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for CS82 (T82a1)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Spinner temperature in °C	1	1	1
Front bearing temperature in °C	2	2	2
Rear bearing temperature in °C	3	3	3
Blade A pitching heat sink temperature in °C	4	4	4
Blade B pitching heat sink temperature in °C	5	5	5
Blade C pitching heat sink temperature in °C	6	6	6
Blade A pitch control box temperature in °C	7	7	7
Blade B pitch control box temperature in °C	8	8	8
Blade C pitch control box temperature in °C	9	9	9
Blade A temperature in °C	10	10	10
Blade B temperature in °C	11	11	11

Description	Raw	Rep	Day
Blade C temperature in °C	12	12	12
Rotor temperature 1 in °C	13	13	13
Rotor temperature 2 in °C	14	14	14
Stator temperature 1 in °C	15	15	15
Stator temperature 2 in °C	16	16	16
Nacelle ambient temperature in °C	17	17	17
Nacelle temperature in °C	18	18	18
Nacelle control cabinet temperature in °C	19	19	19
Rectifier heat sink temperature 1 in °C	20	20	20
Rectifier heat sink temperature 2 in °C	21	21	21
Rectifier cabinet temperature in °C	22	22	22
Excitation controller box heat sink temperature in °C	23	23	23
Ambient temperature in °C	24	24	24
Tower temperature in °C	25	25	25
Control cabinet temperature in °C	26	26	26
Transformer temperature in °C	27	27	27

Tab. 22: *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for CS82 (T82a2)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Sys. 1 power cabinet temperature 1 in °C	1	1	1
Sys. 1 power cabinet temperature 2 in °C	2	2	2
Sys. 1 power cabinet temperature 3 in °C	3	3	3
Sys. 1 power cabinet temperature 4 in °C	4	4	4
Sys. 1 power cabinet temperature 5 in °C	5	5	5
Sys. 1 power cabinet temperature 6 in °C	6	6	6
Sys. 1 power cabinet temperature 7 in °C	7	7	7
Sys. 1 power cabinet temperature 8 in °C	8	8	8
Sys. 1 power cabinet temperature 9 in °C	9	9	9
Sys. 1 power cabinet temperature 10 in °C	10	10	10
Sys. 1 power cabinet temperature 11 in °C	11	11	11

Tab. 23: *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for CS101 (T101a1)

Description	Raw	Rep
Number of sampling values	0	0
Spinner temperature in °C	1	1
Front bearing temperature in °C	2	2

Description	Raw	Rep
Rear bearing temperature in °C	3	3
Blade A pitch control box temperature in °C	4	4
Blade B pitch control box temperature in °C	5	5
Blade C pitch control box temperature in °C	6	6
Blade A temperature in °C	7	7
Blade B temperature in °C	8	8
Blade C temperature in °C	9	9
Rotor temperature 1 in °C	10	10
Rotor temperature 2 in °C	11	11
Stator temperature 1 in °C	12	12
Stator temperature 2 in °C	13	13
Nacelle ambient temperature 1 in °C	14	14
Nacelle ambient temperature 2 in °C	15	15
Nacelle temperature in °C	16	16
Nacelle control cabinet temperature in °C	17	17
Main carrier temperature in °C	18	18
Rectifier cabinet temperature in °C	19	19
Yaw inverter cabinet temperature in °C	20	20
Nacelle fan inverter temperature in °C	21	21
Ambient temperature in °C	22	22
Tower temperature in °C	23	23
Control cabinet temperature in °C	24	24
Transformer temperature in °C	25	25

Tab. 24: *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for CS101 (T101a2)

Description	Raw	Rep
Number of sampling values	0	0
Sys. 1 power cabinet temperature 1 in °C	1	1
Sys. 1 power cabinet temperature 2 in °C	2	2
Sys. 1 power cabinet temperature 3 in °C	3	3
Sys. 1 power cabinet temperature 4 in °C	4	4
Sys. 1 power cabinet temperature 5 in °C	5	5
Sys. 1 power cabinet temperature 6 in °C	6	6
Sys. 1 power cabinet temperature 7 in °C	7	7
Sys. 2 power cabinet temperature 1 in °C	8	8
Sys. 2 power cabinet temperature 2 in °C	9	9

Description	Raw	Rep
Sys. 2 power cabinet temperature 3 in °C	10	10
Sys. 2 power cabinet temperature 4 in °C	11	11
Sys. 2 power cabinet temperature 5 in °C	12	12
Sys. 2 power cabinet temperature 6 in °C	13	13
Sys. 2 power cabinet temperature 7 in °C	14	14

Tab. 25: Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS126 (T126a1)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Front bearing temperature in °C	1	1	1
Rear bearing temperature in °C	2	2	2
Blade A pitch control box temperature in °C	3	3	3
Blade B pitch control box temperature in °C	4	4	4
Blade C pitch control box temperature in °C	5	5	5
Blade A temperature 1 in °C	6	6	6
Blade A temperature 2 in °C	7	7	7
Blade B temperature 1 in °C	8	8	8
Blade B temperature 2 in °C	9	9	9
Blade C temperature 1 in °C	10	10	10
Blade C temperature 2 in °C	11	11	11
Rotor temperature 1 in °C	12	12	12
Rotor temperature 2 in °C	13	13	13
Stator temperature 1.1 in °C	14	14	14
Stator temperature 2.1 in °C	15	15	15
Rectifier cabinet temperature 1.1 in °C	16	16	16
Rectifier cabinet temperature 1.2 in °C	17	17	17
Rectifier cabinet temperature 2.1 in °C	18	18	18
Rectifier cabinet temperature 2.2 in °C	19	19	19
Rectifier cabinet temperature 3.1 in °C	20	20	20
Rectifier cabinet temperature 3.2 in °C	21	21	21
Rectifier cabinet temperature 4.1 in °C	22	22	22
Rectifier cabinet temperature 4.2 in °C	23	23	23
Nacelle ambient temperature 1 in °C	24	24	24
Nacelle temperature in °C	25	25	25
Nacelle control cabinet temperature in °C	26	26	26
Excitation controller box temperature in °C	27	27	27

Description	Raw	Rep	Day
Yaw inverter cabinet temperature in °C	28	28	28
Nacelle fan inverter temperature in °C	29	29	29
Nacelle ambient temperature 2 in °C	30	30	30
Ambient temperature in °C	31	31	31
Tower temperature in °C	32	32	32
Control cabinet temperature in °C	33	33	33
Transformer system temperature 1 in °C	34	34	34
Transformer system temperature 2 in °C	35	35	35
Transformer system temperature 3 in °C	36	36	36
Transformer system temperature 4 in °C	37	37	37

Tab. 26: *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for CS126 (T126a2)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Sys. 1 power cabinet temperature 1 in °C	1	1	1
Sys. 1 power cabinet temperature 2 in °C	2	2	2
Sys. 1 power cabinet temperature 3 in °C	3	3	3
Sys. 1 power cabinet temperature 4 in °C	4	4	4
Sys. 1 power cabinet temperature 5 in °C	5	5	5
Sys. 1 power cabinet temperature 6 in °C	6	6	6
Sys. 1 power cabinet temperature 7 in °C	7	7	7
Sys. 1 power cabinet temperature 8 in °C	8	8	8
Sys. 2 power cabinet temperature 1 in °C	9	9	9
Sys. 2 power cabinet temperature 2 in °C	10	10	10
Sys. 2 power cabinet temperature 3 in °C	11	11	11
Sys. 2 power cabinet temperature 4 in °C	12	12	12
Sys. 2 power cabinet temperature 5 in °C	13	13	13
Sys. 2 power cabinet temperature 6 in °C	14	14	14
Sys. 2 power cabinet temperature 7 in °C	15	15	15
Sys. 2 power cabinet temperature 8 in °C	16	16	16

Tab. 27: *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for CS126 (T126a3)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Sys. 3 power cabinet temperature 1 in °C	1	1	1
Sys. 3 power cabinet temperature 2 in °C	2	2	2
Sys. 3 power cabinet temperature 3 in °C	3	3	3

Description	Raw	Rep	Day
Sys. 3 power cabinet temperature 4 in °C	4	4	4
Sys. 3 power cabinet temperature 5 in °C	5	5	5
Sys. 3 power cabinet temperature 6 in °C	6	6	6
Sys. 3 power cabinet temperature 7 in °C	7	7	7
Sys. 3 power cabinet temperature 8 in °C	8	8	8
Sys. 4 power cabinet temperature 1 in °C	9	9	9
Sys. 4 power cabinet temperature 2 in °C	10	10	10
Sys. 4 power cabinet temperature 3 in °C	11	11	11
Sys. 4 power cabinet temperature 4 in °C	12	12	12
Sys. 4 power cabinet temperature 5 in °C	13	13	13
Sys. 4 power cabinet temperature 6 in °C	14	14	14
Sys. 4 power cabinet temperature 7 in °C	15	15	15
Sys. 4 power cabinet temperature 8 in °C	16	16	16

Tab. 28: The *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for EP3-CS-02 (Tep3c021)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Spinner temperature in °C	1	1	1
Front bearing temperature in °C	2	2	2
Rear bearing temperature in °C	3	3	3
Electrical cabinet A temperature in °C	4	4	4
Electrical cabinet B temperature in °C	5	5	5
Electrical cabinet C temperature in °C	6	6	6
Blade A temperature in °C	7	7	7
Blade B temperature in °C	8	8	8
Blade C temperature in °C	9	9	9
Rotor temperature 1 in °C	10	10	10
Rotor temperature 2 in °C	11	11	11
Cooling-water temperature in °C	12	12	12
Outside temperature 1 (at hub height) in °C	13	13	13
Nacelle temperature in °C	14	14	14
Nacelle control-cabinet temperature in °C	15	15	15
Main-carrier temperature in °C	16	16	16
Yaw inverter control-cabinet temperature in °C	17	17	17
Nacelle fan inverter control-cabinet temperature in °C	18	18	18
Outside temperature (ground) in °C	19	19	19

Description	Raw	Rep	Day
Tower temperature in °C	20	20	20
Control-cabinet temperature in °C	21	21	21
Transformer temperature in °C	22	22	22

Tab. 29: The *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for EP3-CS-02 (Tep3c022)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Converter 1 cabinet temperature in °C	1	1	1
Converter 2 cabinet temperature in °C	2	2	2
Converter 3 cabinet temperature in °C	3	3	3
Converter 4 cabinet temperature in °C	4	4	4
Converter 5 cabinet temperature in °C	5	5	5
Converter 6 cabinet temperature in °C	6	6	6
Converter 7 cabinet temperature in °C	7	7	7
Converter 8 cabinet temperature in °C	8	8	8
Converter 9 cabinet temperature in °C	9	9	9
Converter 10 cabinet temperature in °C	10	10	10
Converter 11 cabinet temperature in °C	11	11	11
Converter 12 cabinet temperature in °C	12	12	12
Converter 13 cabinet temperature in °C	13	13	13
Converter 14 cabinet temperature in °C	14	14	14
Converter 15 cabinet temperature in °C	15	15	15
Converter 16 cabinet temperature in °C	16	16	16

Tab. 30: *Loc\Wec\Plant<N>\Log\T<XX>\<Period>* node for EP4-CS-01 (Tep4c011)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Rotor head temperature in °C	1	1	1
Front rotor bearing temperature in °C	2	2	2
Rear rotor bearing temperature in °C	3	3	3
Blade A pitch control box temperature in °C	4	4	4
Blade B pitch control box temperature in °C	5	5	5
Blade C pitch control box temperature in °C	6	6	6
Blade A temperature in °C	7	7	7
Blade B temperature in °C	8	8	8
Blade C temperature in °C	9	9	9
Rotor temperature 1 in °C	10	10	10

Description	Raw	Rep	Day
Rotor temperature 2 in °C	11	11	11
Stator winding-overhang temperature, front, in °C	12	12	12
Cooling water temperature	13	13	13
Rectifier cabinet temperature 1.1 in °C	14	14	14
Rectifier cabinet temperature 1.2 in °C	15	15	15
Rectifier cabinet temperature 2.1 in °C	16	16	16
Rectifier cabinet temperature 2.2 in °C	17	17	17
Rectifier cabinet temperature 3.1 in °C	18	18	18
Rectifier cabinet temperature 3.2 in °C	19	19	19
Rectifier cabinet temperature 4.1 in °C	20	20	20
Rectifier cabinet temperature 4.2 in °C	21	21	21
Nacelle ambient temperature in °C	22	22	22
Nacelle temperature in °C	23	23	23
Nacelle control cabinet temperature °C	24	24	24
Main carrier temperature in °C	25	25	25
Yaw inverter cabinet temperature in °C	26	26	26
Nacelle fan inverter temperature in °C	27	27	27
Ambient temperature in °C	28	28	28
Tower temperature in °C	29	29	29
Control cabinet temperature in °C	30	30	30
Transformer system temperature 1 in °C	31	31	31
Transformer system temperature 2 in °C	32	32	32

Tab. 31: Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for EP4-CS-01 (Tep4c012)

Description	Raw	Rep	Day
Number of sampling values	0	0	0
Inverter cabinet temperature 1.1 in °C	1	1	1
Inverter cabinet temperature 1.2 in °C	2	2	2
Inverter cabinet temperature 1.3 in °C	3	3	3
Inverter cabinet temperature 1.4 in °C	4	4	4
Inverter cabinet temperature 1.5 in °C	5	5	5
Inverter cabinet temperature 1.6 in °C	6	6	6
Inverter cabinet temperature 1.7 in °C	7	7	7
Inverter cabinet temperature 1.8 in °C	8	8	8
Inverter cabinet temperature 1.9 in °C	9	9	9
Inverter cabinet temperature 1.10 in °C	10	10	10

Description	Raw	Rep	Day
Inverter cabinet temperature 2.1 in °C	11	11	11
Inverter cabinet temperature 2.2 in °C	12	12	12
Inverter cabinet temperature 2.3 in °C	13	13	13
Inverter cabinet temperature 2.4 in °C	14	14	14
Inverter cabinet temperature 2.5 in °C	15	15	15
Inverter cabinet temperature 2.6 in °C	16	16	16
Inverter cabinet temperature 2.7 in °C	17	17	17
Inverter cabinet temperature 2.8 in °C	18	18	18
Inverter cabinet temperature 2.9 in °C	19	19	19
Inverter cabinet temperature 2.10 in °C	20	20	20

3.2.3.5 Loc\Wec\Plant<N>\Log\Wecstd node

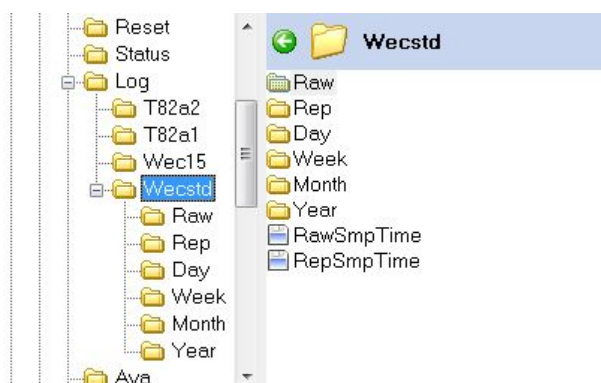


Fig. 9: Loc\Wec\Plant<N>\Log\Wecstd node

The *Wecstd* node contains standard data for the wind energy converter.

Tab. 32: Loc\Wec\Plant<N>\Log\Wecstd node

Node/Item	Description
Raw	Raw data (precalculated WEC data)
Rep	Report data
Day	Daily reports
Week	Weekly reports
Month	Monthly reports
Year	Annual reports
RawSmpTime	Raw data sampling rate in s (default: 60 s)
RepSmpTime	Report data sampling rate in s (default: 600 s)

3.2.3.6 Loc\Wec\Plant<N>\Log\Wecstd\<Period> node

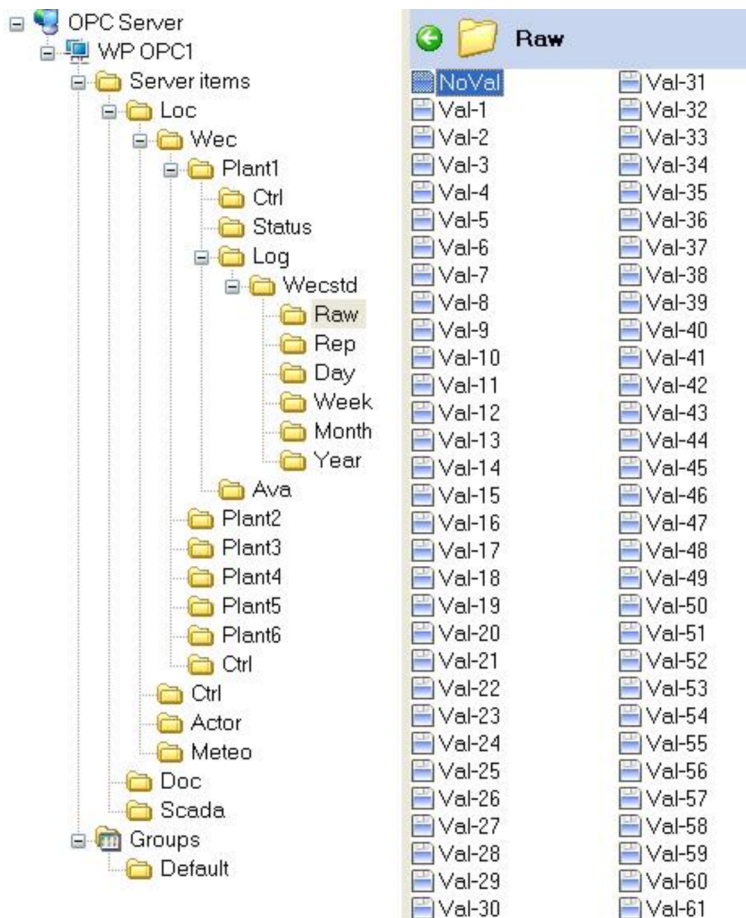


Fig. 10: Loc\Wec\Plant<N>\Log\Wecstd\<Period> node

The following is an explanation of the mean values for one minute. The structure of 10-minute, daily, weekly, monthly and annual report data is identical.

The *Raw* node displays a list of the most recently computed mean values for one minute and their history. The first item *NoVal* shows the maximum number of historical values. This maximum number depends on the selected period (1 minute, 10 minutes, etc.) and can be looked up under *NoVal* for each value.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 33: Loc\Wec\Plant<N>\Log\Wecstd\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean wind speed in m/s	1	1	1	1	1	1
Maximum wind speed in m/s	2	2	2	2	2	2
Minimum wind speed in m/s	3	3	3			
Mean rotor speed in rpm	4	4	4	3	3	3
Maximum rotor speed in rpm	5	5	5	4	4	4

Description	Raw	Rep	Day	Week	Month	Year
Minimum rotor speed in rpm	6	6	6			
Mean power in kW	7	7	7	5	5	5
Maximum power in kW	8	8	8	6	6	6
Minimum power in kW	9	9	9			
Nacelle position in °	10	10	10			
Operating hours in h	11	11	11	7	7	7
Energy produced in kWh	12	12	12	8	8	8
Production time in min	13	13	13	9	9	9
Mean reactive power in kVAr	14	14	14	10	10	10
Maximum reactive power in kVAr	15	15	15	11	11	11
Minimum reactive power in kVAr	16	16	16	12	12	12
Mean value for theoretically available active power in the wind in kW ¹	17	17	17	13	13	13
Mean value for technically available active power in kW ¹	18	18	18	14	14	14
Mean value for maximum available active power limited by force majeure in kW ¹	19	19	19	15	15	15
Mean value for available active power limited by external setpoints in kW ¹	20	20	20	16	16	16
Mean value for blade angle across A, B, C in °	21	21	21			
Mean precipitation in mm/min	22	22	22			
Maximum precipitation in mm/min	23	23	23			
Minimum precipitation in mm/min	24	24	24			
Mean visibility in km ²	25	25	25			
Maximum visibility in km ²	26	26	26			
Minimum visibility in km ²	27	27	27			
Mean ambient brightness in lux ³	28	28	28			
Mean value for Labko ice amplitude for ice detection in % (100 % = No icing; 0 % = Completely iced)	29	29	29			
Mean value of the ice-formation timer in °Cmin	30	30	30			
Mean value of the air pressure in mbar	31	31	31			
Mean value of the air humidity in %	32	32	32			

¹ For a more detailed description, see ch. 3.2, p. 10.

² Only available if visibility sensor is installed.

³ Only available if ambient-brightness sensor is installed (values: 300 day; 0 night).

3.2.3.7 The Loc\Wec\Plant<N>\Log\Weceolog node (optional)

The data from the eologix ice detection system is displayed in the optional *Loc\Wec\Plant<N>\Log\Weceolog* node. The data is only available if the eologix ice detection system is installed and ENERCON enables the data.

Tab. 34: The *Loc\Wec\Plant<N>\Log\Weceolog* node (optional)

Node/item	Description
Raw	Raw data (pre-calculated data)
Rep	Report data
RawSmpTime	Raw-data sampling rate in s (default: 60 s)
RepSmpTime	Report-data sampling rate in s (default: 600 s)

3.2.3.8 The Loc\Wec\Plant<N>\Log\Weceolog\<Period> node (optional)

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 35: The *Loc\Wec\Plant<N>\Log\Weceolog\<Period>* node (optional)

Description	Raw	Rep
Number of sampling values	0	0
Sensor temperature 1 in °C	1	1
Sensor temperature 2 in °C	2	2
Sensor temperature 3 in °C	3	3
Sensor temperature 4 in °C	4	4
Sensor temperature 5 in °C	5	5
Sensor temperature 6 in °C	6	6
Sensor temperature 7 in °C	7	7
Sensor temperature 8 in °C	8	8
Sensor temperature 9 in °C	9	9
Sensor temperature 10 in °C	10	10
Sensor temperature 11 in °C	11	11
Sensor temperature 12 in °C	12	12
Sensor temperature 13 in °C	13	13
Sensor temperature 14 in °C	14	14
Sensor temperature 15 in °C	15	15
Sensor temperature 16 in °C	16	16
Sensor temperature 17 in °C	17	17
Sensor temperature 18 in °C	18	18
Sensor temperature 19 in °C	19	19

Description	Raw	Rep
Sensor temperature 20 in °C	20	20
Sensor temperature 21 in °C	21	21
Sensor temperature 22 in °C	22	22
Sensor temperature 23 in °C	23	23
Sensor temperature 24 in °C	24	24
Sensor temperature 25 in °C	25	25
Sensor temperature 26 in °C	26	26
Sensor temperature 27 in °C	27	27
Sensor temperature 28 in °C	28	28
Sensor temperature 29 in °C	29	29
Sensor temperature 30 in °C	30	30
Sensor temperature 31 in °C	31	31
Sensor temperature 32 in °C	32	32
Sensor temperature 33 in °C	33	33
Sensor temperature 34 in °C	34	34
Sensor temperature 35 in °C	35	35
Sensor temperature 36 in °C	36	36
Sensor temperature 37 in °C	37	37
Sensor temperature 38 in °C	38	38
Sensor temperature 39 in °C	39	39
Sensor temperature 40 in °C	40	40
Sensor temperature 41 in °C	41	41
Sensor temperature 42 in °C	42	42
Sensor temperature 43 in °C	43	43
Sensor temperature 44 in °C	44	44
Sensor temperature 45 in °C	45	45
Sensor temperature 46 in °C	46	46
Sensor temperature 47 in °C	47	47
Sensor temperature 48 in °C	48	48
Sensor temperature 49 in °C	49	49
Sensor temperature 50 in °C	50	50
Icing level, sensor 1	51	51
Icing level, sensor 2	52	52
Icing level, sensor 3	53	53
Icing level, sensor 4	54	54
Icing level, sensor 5	55	55

Description	Raw	Rep
Icing level, sensor 6	56	56
Icing level, sensor 7	57	57
Icing level, sensor 8	58	58
Icing level, sensor 9	59	59
Icing level, sensor 10	60	60
Icing level, sensor 11	61	61
Icing level, sensor 12	62	62
Icing level, sensor 13	63	63
Icing level, sensor 14	64	64
Icing level, sensor 15	65	65
Icing level, sensor 16	66	66
Icing level, sensor 17	67	67
Icing level, sensor 18	68	68
Icing level, sensor 19	69	69
Icing level, sensor 20	70	70
Icing level, sensor 21	71	71
Icing level, sensor 22	72	72
Icing level, sensor 23	73	73
Icing level, sensor 24	74	74
Icing level, sensor 25	75	75
Icing level, sensor 26	76	76
Icing level, sensor 27	77	77
Icing level, sensor 28	78	78
Icing level, sensor 29	79	79
Icing level, sensor 30	80	80
Icing level, sensor 31	81	81
Icing level, sensor 32	82	82
Icing level, sensor 33	83	83
Icing level, sensor 34	84	84
Icing level, sensor 35	85	85
Icing level, sensor 36	86	86
Icing level, sensor 37	87	87
Icing level, sensor 38	88	88
Icing level, sensor 39	89	89
Icing level, sensor 40	90	90
Icing level, sensor 41	91	91

Description	Raw	Rep
Icing level, sensor 42	92	92
Icing level, sensor 43	93	93
Icing level, sensor 44	94	94
Icing level, sensor 45	95	95
Icing level, sensor 46	96	96
Icing level, sensor 47	97	97
Icing level, sensor 48	98	98
Icing level, sensor 49	99	99
Icing level, sensor 50	100	100
Electrical-cabinet temperature in °C	101	101
Processor temperature in °C	102	102

3.2.3.9 The Loc\Wec\Plant<N>\Log\Wecfos4x node (optional)

The data from the fos4X ice detection system is displayed in the optional *Loc\Wec\Plant<N>\Log\Wecfos4x* node. The data is only available if the fos4X ice detection system is installed and ENERCON enables the data.

Tab. 36: The Loc\Wec\Plant<N>\Log\Wecfos4x node (optional)

Node/item	Description
Raw	Raw data (pre-calculated data)
Rep	Report data
RawSmpTime	Raw-data sampling rate in s (default: 60 s)
RepSmpTime	Report-data sampling rate in s (default: 600 s)

3.2.3.10 The Loc\Wec\Plant<N>\Log\Wecfos4x\<Period> node (optional)

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 37: The Loc\Wec\Plant<N>\Log\Wecfos4x\<Period> node (optional)

Description	Raw	Rep
Number of sampling values	0	0
Ice weight on rotor blade A in kg	1	1
Ice weight on rotor blade B in kg	2	2
Ice weight on rotor blade C in kg	3	3
Predicted ice weight on rotor blade A in kg	4	4
Predicted ice weight on rotor blade B in kg	5	5

Description	Raw	Rep
Predicted ice weight on rotor blade C in kg	6	6
Electrical-cabinet temperature in °C	7	7
Processor temperature in °C	8	8

3.2.3.11 The Loc\Wec\Plant<N>\Log\Wecwoef node (optional)

The data from the Wölfel ice detection system is displayed in the optional *Loc\Wec\Plant<N>\Log\Wecwoef* node. The data is only available if the Wölfel ice detection system is installed and ENERCON enables the data.

Tab. 38: The *Loc\Wec\Plant<N>\Log\Wecwoef* node (optional)

Node/item	Description
Raw	Raw data (pre-calculated data)
Rep	Report data
RawSmpTime	Raw-data sampling rate in s (default: 60 s)
RepSmpTime	Report-data sampling rate in s (default: 600 s)

3.2.3.12 The Loc\Wec\Plant<N>\Log\Wecwoef<Period> node (optional)

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 39: The *Loc\Wec\Plant<N>\Log\Wecwoef<Period>* node (optional)

Description	Raw	Rep
Number of sampling values	0	0
Ice indicator, rotor blade A	1	1
Ice indicator, rotor blade B	2	2
Ice indicator, rotor blade C	3	3
Electrical-cabinet temperature in °C	4	4
Processor temperature in °C	5	5

3.2.4 Loc\Wec\Plant<N>\Ava node

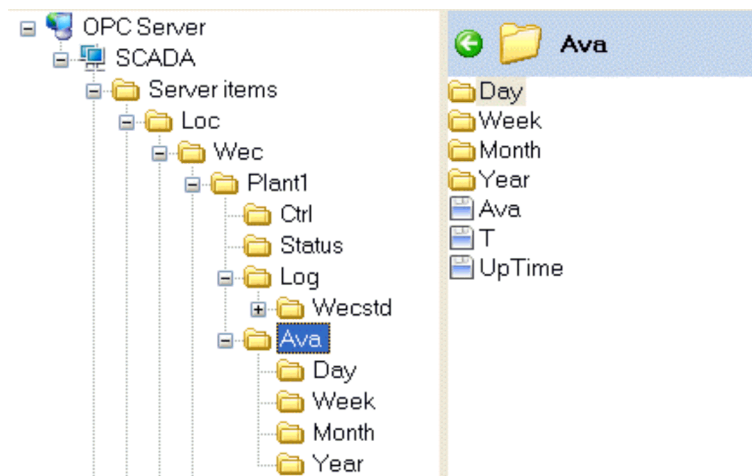


Fig. 11: Loc\Wec\Plant<N>\Ava node

The *Ava* node contains report data about the availability of *Plant<N>* and its history.

This node exists for all plant groups. The following is an explanation of the node for one wind energy converter <N> only.

Tab. 40: Node Loc\Wec\Plant<N>\Ava – Nodes and items

Node/Item	Description
Day	Daily availability
Week	Weekly availability
Month	Monthly availability
Year	Annual availability
Ava	Computed total availability in % since commissioning of plant <N>
T	Array of T times as defined in the ENERCON PartnerKonzept (EPK)
UpTime	Start of computation of absolute availability; typically the commissioning date of plant <N> (date and time)

Tab. 41: Node Loc\Wec\Plant<N>\Ava – Elements of the array

Index	Description
0	T1: Time in seconds during which the plant is in operation or ready for operation, the plant connections are available, no faults are displayed, and the remote monitoring system is working.
1	T2: Total time since the start of availability measurement in seconds
2	T3: Time in seconds during which plant connections are not available. A plant connection is unavailable if the connection parameters are out of range. Parameters in the utility grid are grid voltage and grid frequency at the plant. Parameters are also out of range if grid faults occur that are not caused by the plant.

Index	Description
3	T4: Time in seconds during which the plant is not in operation for the following reasons: ■ The plant has been taken out of operation by the customer or at the customer's request due to modifications, parameter configuration, measurements, maintenance work, or shutdown for operational reasons (except repairs) ■ Force majeure ■ Damage to the plant, for which the manufacturer is not responsible
4	T5: Time in seconds during which the remote monitoring system has failed or computed data have been lost.
5	T6: Time in seconds during which the plant indicates a <i>Fault</i> and is therefore unavailable.

Availability of WECs is computed as follows:

$$\frac{T1}{T2-T3-T4-T5}$$

Fig. 12: T-time formula

3.2.4.1 Loc\Wec\Plant<N>\Ava<Period> node

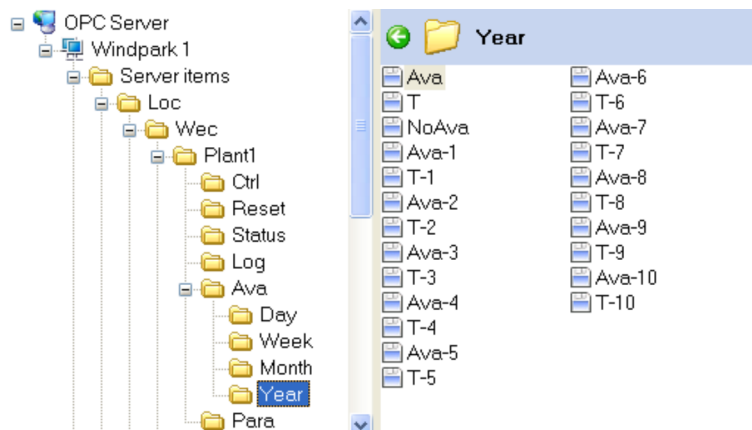


Fig. 13: Log\Wec\Plant<N>\Ava<Period> node

The <Period> node contains availability data for historical time periods.

Tab. 42: Loc\Wec\Plant<N>\Ava<Period> node

Item	Description
Ava	Computed total availability in % since commissioning of plant <N>
T	T-times array as defined in the ENERCON PartnerKonzept (EPK) (current time interval); see also ch. 3.2.4, p. 41
NoAva	Number of data records
Ava-1	Computed total availability in % since commissioning of WEC <N> for the previous time interval
T-1	T-times array as defined in the ENERCON PartnerKonzept (EPK) (current time interval); see also ch. 3.2.4, p. 41

3.2.5 Loc\Wec\Plant<N>\Para node

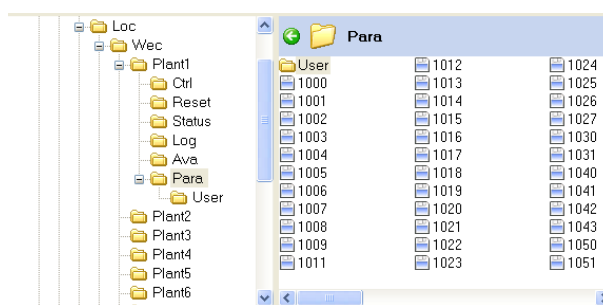


Fig. 14: Loc\Wec\Plant<N>\Para node

The *Para* node can be used to view the control parameters of a wind energy converter. The item name corresponds to the parameter name used in the control system.

The *Loc\Wec\Plant<N>\Para\User* node also includes items whose names correspond to the names of the parameters in the control system. The values of these items indicate which user changed the relevant parameter most recently.

3.3 Loc\Sens node

This node contains the data of the *sensor plants* group.

3.3.1 Loc\Sens\Plant<N> node

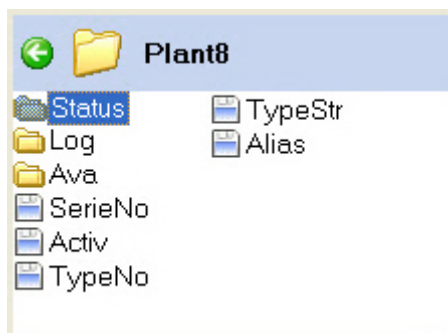


Fig. 15: Loc\Sens\Plant<N> node

This node contains the data of one sensor unit (e.g. medium-voltage data acquisition).

Tab. 43: Loc\Sens\Plant<N> node

Node/item	Description
Status	Status data for this sensor unit
Log	Report data for this sensor unit
Ava	Availability data for this sensor unit
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer

3.3.2 Loc\Sens\Plant<N>\Log\Mspgerf1\<Period> node

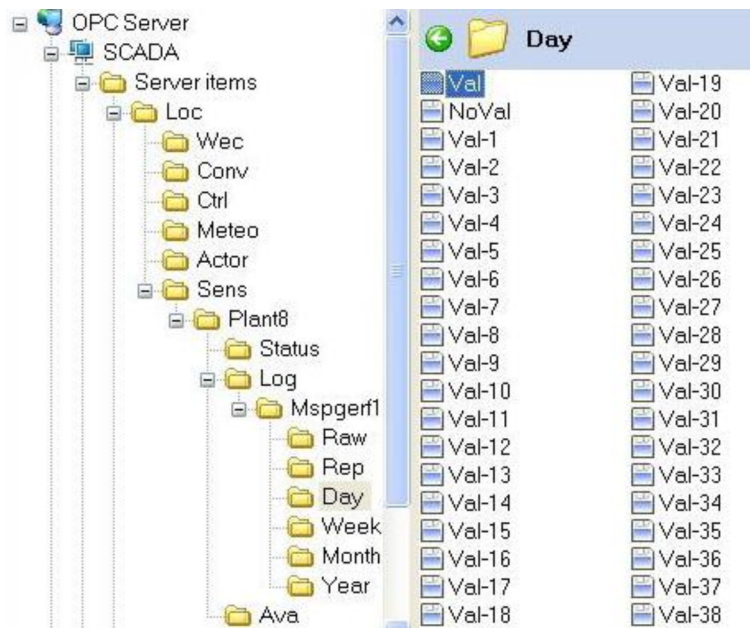


Fig. 16: Loc\Sens\Plant<N>\Log\Mspgerf1\<Period> node

This node contains the report data of the medium-voltage acquisition data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val*-<X> are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 44: Loc\Sens\Plant<N>\Log\Mspgerf1\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean voltage U12 in V	1	1	1	1	1	1
Maximum voltage U12 in V	2	2	2	2	2	2
Minimum voltage U12 in V	3	3	3	3	3	3
Mean voltage U23 in V	4	4	4	4	4	4
Maximum voltage U23 in V	5	5	5	5	5	5
Minimum voltage U23 in V	6	6	6	6	6	6
Mean voltage U31 in V	7	7	7	7	7	7
Maximum voltage U31 in V	8	8	8	8	8	8
Minimum voltage U31 in V	9	9	9	9	9	9
Mean frequency in Hz	10	10	10	10	10	10
Maximum frequency in Hz	11	11	11	11	11	11
Minimum frequency in Hz	12	12	12	12	12	12

3.4 Loc\Conv node

This node contains the data of the *converter plants* group.

3.4.1 Solar converter

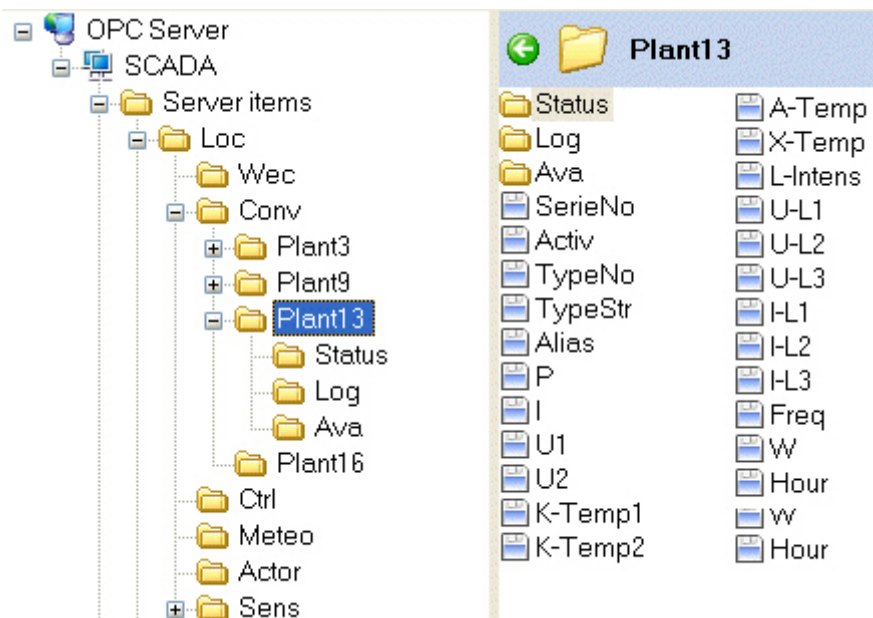


Fig. 17: Loc\Conv\Plant<N> node of an ENERCON solar converter

The nodes and items of an ENERCON solar converter are described below.

Tab. 45: Loc\Conv\Plant<N> node of an ENERCON solar converter

Node/item	Description
Status	Status data of this solar converter
Log	Report data of this solar converter
Ava	Availability data of this solar converter
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
P	Power in kW
I	Current in A
U1	Voltage level 1 in V
U2	Voltage level 2 in V
K-Temp1	-----
K-Temp2	-----
A-Temp	Ambient temperature in °C
X-Temp	Freely selectable temperature in °C

Node/item	Description
L-Intens	-----
U-L1	Low voltage U12 in V
U-L2	Low voltage U23 in V
U-L3	Low voltage U31 in V
I-L1	Power feed L1 in A
I-L2	Power feed L2 in A
I-L3	Power feed L3 in A
Freq	Grid frequency in Hz
W	Energy in kWh
Hour	Operating hours in h

3.4.1.1 Loc\Conv\Plant<N>\Log\Solar1\<Period> node

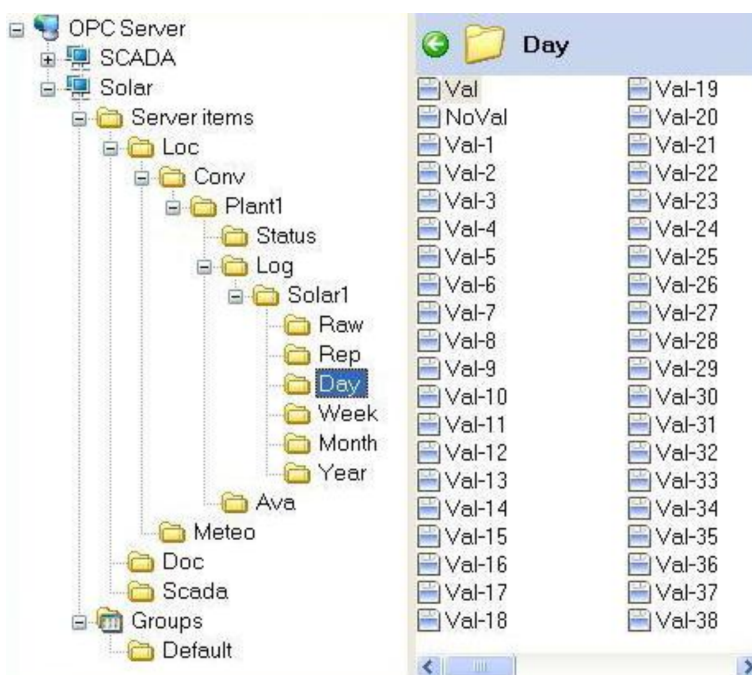


Fig. 18: Loc\Conv\Plant<N>\Log\Solar1\<Period> node

This node contains the report data of the solar converter data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val*-<X> are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 46: The Loc\Conv\Plant<N>\Log\Solar1\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean power in kW	1	1	1	1	1	1
Maximum power in kW	2	2	2	2	2	2

Description	Raw	Rep	Day	Week	Month	Year
Minimum power in kW	3	3	3	3	3	3
Mean current in A	4	4	4	4	4	4
Maximum current in A	5	5	5	5	5	5
Minimum current in A	6	6	6	6	6	6
Mean voltage level 1 in V	7	7	7	7	7	7
Maximum voltage level 1 in V	8	8	8	8	8	8
Minimum voltage level 1 in V	9	9	9	9	9	9
Mean voltage level 2 in V	10	10	10	10	10	10
Maximum voltage level 2 in V	11	11	11	11	11	11
Minimum voltage level 2 in V	12	12	12	12	12	12
Mean solar irradiance in W/m ²	13	13	13	13	13	13
Maximum solar irradiance in W/m ²	14	14	14	14	14	14
Minimum solar irradiance in W/m ²	15	15	15	15	15	15
Mean voltage level U12 in V	16	16	16	16	16	16
Maximum voltage level U12 in V	17	17	17	17	17	17
Minimum voltage level U12 in V	18	18	18	18	18	18
Mean voltage level U23 in V	19	19	19	19	19	19
Maximum voltage level U23 in V	20	20	20	20	20	20
Minimum voltage level U23 in V	21	21	21	21	21	21
Mean voltage level U31 in V	22	22	22	22	22	22
Maximum voltage level U31 in V	23	23	23	23	23	23
Minimum voltage level U31 in V	24	24	24	24	24	24
Mean current L1 in A	25	25	25	25	25	25
Maximum current L1 in A	26	26	26	26	26	26
Minimum current L1 in A	27	27	27	27	27	27
Mean current L2 in A	28	28	28	28	28	28
Maximum current L2 in A	29	29	29	29	29	29
Minimum current L2 in A	30	30	30	30	30	30
Mean current L3 in A	31	31	31	31	31	31
Maximum current L3 in A	32	32	32	32	32	32
Minimum current L3 in A	33	33	33	33	33	33
Mean frequency in Hz	34	34	34	34	34	34
Maximum frequency in Hz	35	35	35	35	35	35
Minimum frequency in Hz	36	36	36	36	36	36
Energy produced in kWh	37	37	37	37	37	37
Operating time in h	38	38	38	38	38	38

3.4.2 Feeding and charging unit

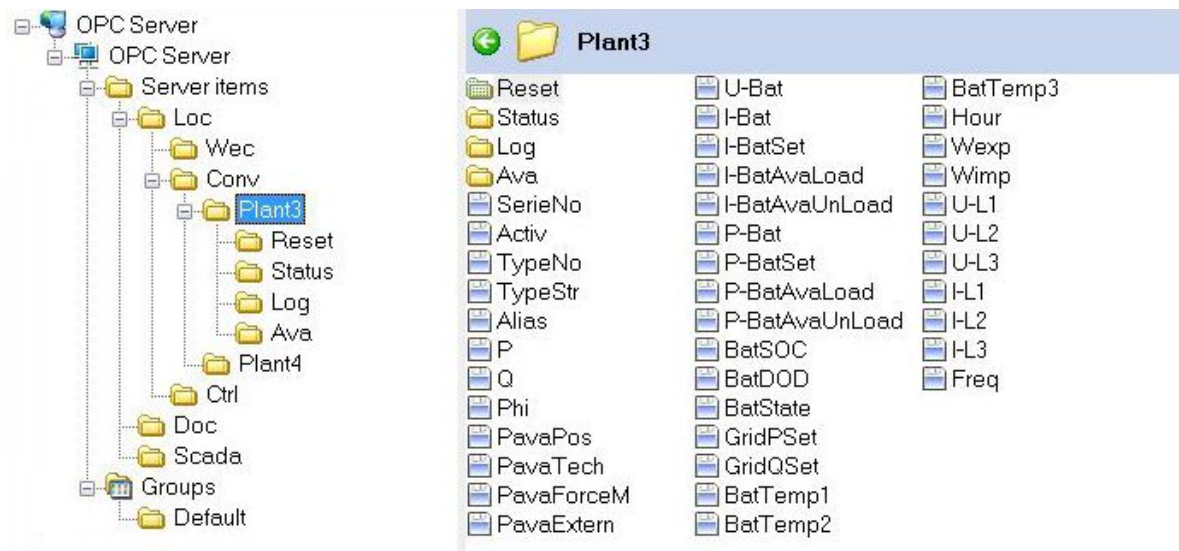


Fig. 19: Loc\Conv\Plant<N> of the ENERCON feeding and charging unit

The nodes and items of an ENERCON feeding and charging unit are described below.

Tab. 47: Loc\Conv\Plant<N> node of the ENERCON feeding and charging unit

Node/item	Description
Reset	Reset of the control system
Status	Status data of the feeding and charging unit
Log	Report data of the feeding and charging unit
Ava	Availability data of the feeding and charging unit
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
P	Power in kW
Q	Reactive power in kVar
Phi	Phase angle in °
PavaPos	Available power of the feeding and charging unit in kW
PavaTech	Technically available power of the feeding and charging unit in kW
PavaForceM	Maximum available power of the feeding and charging unit limited by force majeure in kW
PavaExtern	Maximum available power of the feeding and charging unit limited by external setpoints in kW
U-Bat	Battery voltage in V
I-Bat	Battery current in A
I-BatSet	Battery current setpoint in A

Node/item	Description
I-BatAvaLoad	Battery charge current in A
I-BatAvaUn-load	Battery discharge current in A
BatSOC	Charge status of battery (SoC) in %
BatDOD	Depth of discharge of battery (DoD) in %
BatState	Battery status
GridPSet	Grid power setpoint in kW
GridQSet	Grid reactive power setpoint in kVAr
BatTemp1	Battery temperature 1 in °C
BatTemp2	Battery temperature 2 in °C
BatTemp3	Battery temperature 3 in °C
Hour	Operating hours in h
Wexp	Exported energy in kWh
Wimp	Imported energy in kWh
U-L1	Voltage L1 in V
U-L2	Voltage L2 in V
U-L3	Voltage L3 in V
I-L1	Current L1 in A
I-L2	Current L2 in A
I-L3	Current L3 in A
Freq	Frequency in Hz

3.4.2.1 Loc\Conv\Plant<N>\Log\Bcs1\<Period> node

This node contains the report data of the feeding and charging unit data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 48: Loc\Conv\Plant<N>\Log\Bcs1\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean power in kW	1	1	1	1	1	1
Maximum power in kW	2	2	2	2	2	2
Minimum power in kW	3	3	3	3	3	3
Mean reactive power in kVAr	4	4	4	4	4	4
Maximum reactive power in kVAr	5	5	5	5	5	5

Description	Raw	Rep	Day	Week	Month	Year
Minimum reactive power in kVAr	6	6	6	6	6	6
Mean available power in kW	7	7	7	7	7	7
Mean technically available power in kW	8	8	8	8	8	8
Mean available power limited by force majeure in kW	9	9	9	9	9	9
Mean available power limited by external setpoints in kW	10	10	10	10	10	10
Mean battery voltage in V	11	11	11	11	11	11
Maximum battery voltage in V	12	12	12	12	12	12
Minimum battery voltage in V	13	13	13	13	13	13
Mean battery current in A	14	14	14	14	14	14
Maximum battery current in A	15	15	15	15	15	15
Minimum battery current in A	16	16	16	16	16	16
Mean battery power in kW	17	17	17	17	17	17
Maximum battery power in kW	18	18	18	18	18	18
Minimum battery power in kW	19	19	19	19	19	19
Mean charge status (SoC) in %	20	20	20	20	20	20
Mean depth of discharge (DoD) in %	21	21	21	21	21	21
Mean ambient temperature in °C	22	22	22	22	22	22
Mean control cabinet temperature in °C	23	23	23	23	23	23
Mean transformer system temperature in °C	24	24	24	24	24	24
Mean transformer ambient temperature in °C	25	25	25	25	25	25
Mean low-voltage room temperature 1 in °C	26	26	26	26	26	26
Mean low-voltage room temperature 2 in °C	27	27	27	27	27	27
Mean battery module temperature in °C	28	28	28	28	28	28
Operating hours in h	29	29	29	29	29	29
Energy meter, consumption in kWh	30	30	30	30	30	30
Energy meter, production in kWh	31	31	31	31	31	31
Operating time in min	32	32	32	32	32	32

3.4.2.2 Loc\Conv\Plant<N>\Log\Bcs2\<Period> node

This node contains the report data of the feeding- and charging-unit data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 49: Loc\Conv\Plant<N>\Log\Bcs2\<Period> node

Description	Raw	Rep	Day
Number of sampling values	0	0	0
AC/DC rectifier 1 heat sink L1+ temperature in °C	1	1	1
AC/DC rectifier 1 heat sink L1- temperature in °C	2	2	2
AC/DC rectifier 1 heat sink L2+ temperature in °C	3	3	3
AC/DC rectifier 1 heat sink L2- temperature in °C	4	4	4
AC/DC rectifier 1 heat sink L3+ temperature in °C	5	5	5
AC/DC rectifier 1 heat sink L3- temperature in °C	6	6	6
AC/DC rectifier 1 step-up converter heat sink temperature in °C	7	7	7
AC/DC rectifier 1 chopper heat sink temperature in °C	8	8	8
AC/DC rectifier 1 electrical cabinet heat sink temperature in °C	9	9	9
AC/DC rectifier 2 heat sink L1+ temperature in °C	10	10	10
AC/DC rectifier 2 heat sink L1- temperature in °C	11	11	11
AC/DC rectifier 2 heat sink L2+ temperature in °C	12	12	12
AC/DC rectifier 2 heat sink L2- temperature in °C	13	13	13
AC/DC rectifier 2 heat sink L3+ temperature in °C	14	14	14
AC/DC rectifier 2 heat sink L3- temperature in °C	15	15	15
AC/DC rectifier 2 step-up converter heat sink temperature in °C	16	16	16
AC/DC rectifier 2 chopper heat sink temperature in °C	17	17	17
AC/DC rectifier 2 electrical cabinet heat sink temperature in °C	18	18	18
AC/DC rectifier 3 heat sink L1+ temperature in °C	19	19	19
AC/DC rectifier 3 heat sink L1- temperature in °C	20	20	20
AC/DC rectifier 3 heat sink L2+ temperature in °C	21	21	21
AC/DC rectifier 3 heat sink L2- temperature in °C	22	22	22
AC/DC rectifier 3 heat sink L3+ temperature in °C	23	23	23
AC/DC rectifier 3 heat sink L3- temperature in °C	24	24	24
AC/DC rectifier 3 step-up converter heat sink temperature in °C	25	25	25
AC/DC rectifier 3 chopper heat sink temperature in °C	26	26	26
AC/DC rectifier 3 electrical cabinet heat sink temperature in °C	27	27	27
AC/DC rectifier 4 heat sink L1+ temperature in °C	28	28	28
AC/DC rectifier 4 heat sink L1- temperature in °C	29	29	29
AC/DC rectifier 4 heat sink L2+ temperature in °C	30	30	30
AC/DC rectifier 4 heat sink L2- temperature in °C	31	31	31

Description	Raw	Rep	Day
AC/DC rectifier 4 heat sink L3+ temperature in °C	32	32	32
AC/DC rectifier 4 heat sink L3- temperature in °C	33	33	33
AC/DC rectifier 4 step-up converter heat sink temperature in °C	34	34	34
AC/DC rectifier 4 chopper heat sink temperature in °C	35	35	35
AC/DC rectifier 4 electrical cabinet heat sink temperature in °C	36	36	36
AC/DC rectifier 5 heat sink L1+ temperature in °C	37	37	37
AC/DC rectifier 5 heat sink L1- temperature in °C	38	38	38
AC/DC rectifier 5 heat sink L2+ temperature in °C	39	39	39
AC/DC rectifier 5 heat sink L2- temperature in °C	40	40	40
AC/DC rectifier 5 heat sink L3+ temperature in °C	41	41	41
AC/DC rectifier 5 heat sink L3- temperature in °C	42	42	42
AC/DC rectifier 5 step-up converter heat sink temperature in °C	43	43	43
AC/DC rectifier 5 chopper heat sink temperature in °C	44	44	44
AC/DC rectifier 5 electrical cabinet heat sink temperature in °C	45	45	45
AC/DC rectifier 6 heat sink L1+ temperature in °C	46	46	46
AC/DC rectifier 6 heat sink L1- temperature in °C	47	47	47
AC/DC rectifier 6 heat sink L2+ temperature in °C	48	48	48
AC/DC rectifier 6 heat sink L2- temperature in °C	49	49	49
AC/DC rectifier 6 heat sink L3+ temperature in °C	50	50	50
AC/DC rectifier 6 heat sink L3- temperature in °C	51	51	51
AC/DC rectifier 6 step-up converter heat sink temperature in °C	52	52	52
AC/DC rectifier 6 chopper heat sink temperature in °C	53	53	53
AC/DC rectifier 6 electrical cabinet heat sink temperature in °C	54	54	54
AC/DC rectifier 7 heat sink L1+ temperature in °C	55	55	55
AC/DC rectifier 7 heat sink L1- temperature in °C	56	56	56
AC/DC rectifier 7 heat sink L2+ temperature in °C	57	57	57
AC/DC rectifier 7 heat sink L2- temperature in °C	58	58	58
AC/DC rectifier 7 heat sink L3+ temperature in °C	59	59	59
AC/DC rectifier 7 heat sink L3- temperature in °C	60	60	60
AC/DC rectifier 7 step-up converter heat sink temperature in °C	61	61	61
AC/DC rectifier 7 chopper heat sink temperature in °C	62	62	62
AC/DC rectifier 7 electrical cabinet heat sink temperature in °C	63	63	63
DC-DC converter 1 heat sink L1+ temperature in °C	64	64	64
DC-DC converter 1 heat sink L1- temperature in °C	65	65	65
DC-DC converter 1 heat sink L2+ temperature in °C	66	66	66
DC-DC converter 1 heat sink L2- temperature in °C	67	67	67

Description	Raw	Rep	Day
DC-DC converter 1 heat sink L3+ temperature in °C	68	68	68
DC-DC converter 1 heat sink L3- temperature in °C	69	69	69
DC-DC converter 1 heat sink L4+ temperature in °C	70	70	70
DC-DC converter 1 heat sink L4- temperature in °C	71	71	71
DC-DC converter 1 electrical cabinet heat sink temperature in °C	72	72	72
DC-DC converter 2 heat sink L1+ temperature in °C	73	73	73
DC-DC converter 2 heat sink L1- temperature in °C	74	74	74
DC-DC converter 2 heat sink L2+ temperature in °C	75	75	75
DC-DC converter 2 heat sink L2- temperature in °C	76	76	76
DC-DC converter 2 heat sink L3+ temperature in °C	77	77	77
DC-DC converter 2 heat sink L3- temperature in °C	78	78	78
DC-DC converter 2 heat sink L4+ temperature in °C	79	79	79
DC-DC converter 2 heat sink L4- temperature in °C	80	80	80
DC-DC converter 2 electrical cabinet heat sink temperature in °C	81	81	81
DC-DC converter 3 heat sink L1+ temperature in °C	82	82	82
DC-DC converter 3 heat sink L1- temperature in °C	83	83	83
DC-DC converter 3 heat sink L2+ temperature in °C	84	84	84
DC-DC converter 3 heat sink L2- temperature in °C	85	85	85
DC-DC converter 3 heat sink L3+ temperature in °C	86	86	86
DC-DC converter 3 heat sink L3- temperature in °C	87	87	87
DC-DC converter 3 heat sink L4+ temperature in °C	88	88	88
DC-DC converter 3 heat sink L4- temperature in °C	89	89	89
DC-DC converter 3 electrical cabinet heat sink temperature in °C	90	90	90
DC-DC converter 4 heat sink L1+ temperature in °C	91	91	91
DC-DC converter 4 heat sink L1- temperature in °C	92	92	92
DC-DC converter 4 heat sink L2+ temperature in °C	93	93	93
DC-DC converter 4 heat sink L2- temperature in °C	94	94	94
DC-DC converter 4 heat sink L3+ temperature in °C	95	95	95
DC-DC converter 4 heat sink L3- temperature in °C	96	96	96
DC-DC converter 4 heat sink L4+ temperature in °C	97	97	97
DC-DC converter 4 heat sink L4- temperature in °C	98	98	98
DC-DC converter 4 electrical cabinet heat sink temperature in °C	99	99	99
DC-DC converter 5 heat sink L1+ temperature in °C	100	100	100
DC-DC converter 5 heat sink L1- temperature in °C	101	101	101
DC-DC converter 5 heat sink L2+ temperature in °C	102	102	102
DC-DC converter 5 heat sink L2- temperature in °C	103	103	103

Description	Raw	Rep	Day
DC-DC converter 5 heat sink L3+ temperature in °C	104	104	104
DC-DC converter 5 heat sink L3- temperature in °C	105	105	105
DC-DC converter 5 heat sink L4+ temperature in °C	106	106	106
DC-DC converter 5 heat sink L4- temperature in °C	107	107	107
DC-DC converter 5 electrical cabinet heat sink temperature in °C	108	108	108
DC-DC converter 6 heat sink L1+ temperature in °C	109	109	109
DC-DC converter 6 heat sink L1- temperature in °C	110	110	110
DC-DC converter 6 heat sink L2+ temperature in °C	111	111	111
DC-DC converter 6 heat sink L2- temperature in °C	112	112	112
DC-DC converter 6 heat sink L3+ temperature in °C	113	113	113
DC-DC converter 6 heat sink L3- temperature in °C	114	114	114
DC-DC converter 6 heat sink L4+ temperature in °C	115	115	115
DC-DC converter 6 heat sink L4- temperature in °C	116	116	116
DC-DC converter 6 electrical cabinet heat sink temperature in °C	117	117	117
DC-DC converter 7 heat sink L1+ temperature in °C	118	118	118
DC-DC converter 7 heat sink L1- temperature in °C	119	119	119
DC-DC converter 7 heat sink L2+ temperature in °C	120	120	120
DC-DC converter 7 heat sink L2- temperature in °C	121	121	121
DC-DC converter 7 heat sink L3+ temperature in °C	122	122	122
DC-DC converter 7 heat sink L3- temperature in °C	123	123	123
DC-DC converter 7 heat sink L4+ temperature in °C	124	124	124
DC-DC converter 7 heat sink L4- temperature in °C	125	125	125
DC-DC converter 7 electrical cabinet heat sink temperature in °C	126	126	126

3.5 Loc\Ctrl node

This node contains the data of the *control plants* group.

3.5.1 RTU

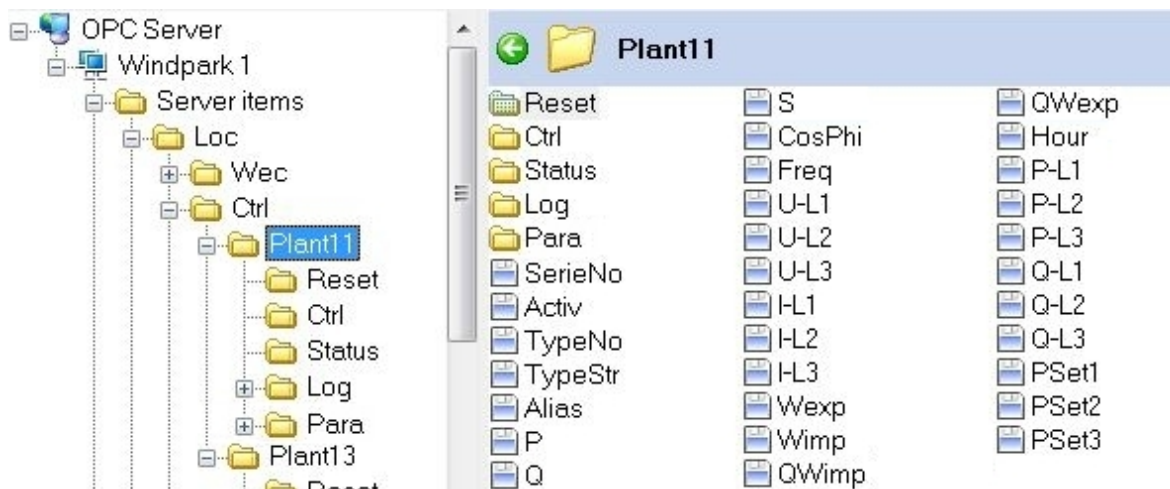


Fig. 20: Loc\Ctrl\Plant<N> node of the RTU

The nodes and items of the ENERCON SCADA Remote Terminal Unit (RTU) are described below.

Tab. 50: Loc\Ctrl\Plant<N> node of the RTU

Node/item	Description
Reset	Reset of the control system
Ctrl	Items for RTU control
Status	RTU status data
Log	RTU report data
Para	RTU parameters
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
P	Power in kW
Q	Reactive power in kVAr
S	Apparent power in VA
CosPhi	Power factor Cos Phi
Freq	Frequency in Hz
U-L1	Voltage U12 in V
U-L2	Voltage U23 in V
U-L3	Voltage U31 in V

Node/item	Description
I-L1	Current L1 in A
I-L2	Current L2 in A
I-L3	Current L3 in A
Wexp	Exported energy in kWh
Wimp	Imported energy in kWh
QWimp	Reactive power during energy import in kVArh
QWexp	Reactive power during energy export in kVArh
Hour	Operating hours in h
P-L1	Active power L1 in kW
P-L2	Active power L2 in kW
P-L3	Active power L3 in kW
Q-L1	Reactive power L1 in kVA
Q-L2	Reactive power L2 in kVA
Q-L3	Reactive power L3 in kVA
PSet1	Active power setpoint 1 in kW
PSet2	Active power setpoint 2 in kW
PSet3	Active power setpoint 3 in kW

3.5.1.1 Loc\Ctrl\Plant<N>\Log\Rtu1\<Period> node

This node contains the report data of the RTU data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 51: Loc\Ctrl\Plant<N>\Log\Rtu1\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean power in kW	1	1	1	1	1	1
Maximum power in kW	2	2	2	2	2	2
Minimum power in kW	3	3	3	3	3	3
Mean reactive power in kVAr	4	4	4	4	4	4
Maximum reactive power in kVAr	5	5	5	5	5	5
Minimum reactive power in kVAr	6	6	6	6	6	6
Mean apparent power in VA	7	7	7	7	7	7
Maximum apparent power in VA	8	8	8	8	8	8
Minimum apparent power in VA	9	9	9	9	9	9
Mean Cos Phi	10	10	10	10	10	10
Maximum Cos Phi	11	11	11	11	11	11
Minimum Cos Phi	12	12	12	12	12	12
Mean frequency in Hz	13	13	13	13	13	13
Maximum frequency in Hz	14	14	14	14	14	14
Minimum frequency in Hz	15	15	15	15	15	15
Mean voltage U12 in V	16	16	16	16	16	16
Maximum voltage U12 in V	17	17	17	17	17	17
Minimum voltage U12 in V	18	18	18	18	18	18
Mean voltage U23 in V	19	19	19	19	19	19
Maximum voltage U23 in V	20	20	20	20	20	20
Minimum voltage U23 in V	21	21	21	21	21	21
Mean voltage U31 in V	22	22	22	22	22	22
Maximum voltage U31 in V	23	23	23	23	23	23
Minimum voltage U31 in V	24	24	24	24	24	24
Mean current 1 L1 in A	25	25	25	25	25	25
Maximum current 1 L1 in A	26	26	26	26	26	26
Minimum current 1 L1 in A	27	27	27	27	27	27
Mean current 1 L2 in A	28	28	28	28	28	28

Description	Raw	Rep	Day	Week	Month	Year
Maximum current 1 L2 in A	29	29	29	29	29	29
Minimum current 1 L2 in A	30	30	30	30	30	30
Mean current 1 L3 in A	31	31	31	31	31	31
Maximum current 1 L3 in A	32	32	32	32	32	32
Minimum current 1 L3 in A	33	33	33	33	33	33
Energy meter, production in kWh	34	34	34	34	34	34
Energy meter, consumption in kWh	35	35	35	35	35	35
Reactive power meter (inductive) in kVAr	36	36	36	36	36	36
Reactive power meter (capacitive) in kVAr	37	37	37	37	37	37
Operating hours in h	38	38	38	38	38	38
Mean active power setpoint 1 in kW	39	39	39	39	39	39
Mean active power setpoint 2 in kW	40	40	40	40	40	40
Mean active power setpoint 3 in kW	41	41	41	41	41	41

3.5.1.2 Loc\Ctrl\Plant<N>\Log\Rtu15\<Period> node (optional)

This optional node contains the report data of the RTU data logger as 15-minute data for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 52: Loc\Ctrl\Plant<N>\Log\Rtu15\<Period> node (optional)

Description	Raw	Rep
Number of sampling values	0	0
Mean power in kW	1	1
Mean reactive power in kVAr	2	2
Mean apparent power in VA	3	3
Energy meter, production in kWh	4	4
Energy meter, consumption in kWh	5	5
Reactive power meter (inductive) in kVAr	6	6
Reactive power meter (capacitive) in kVAr	7	7
Operating hours in h	8	8

3.5.1.3 Loc\Ctrl\Plant<N>\Log\Rtuphase1\<Period> node

This node contains the report data of the RTU data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 53: Loc\Ctrl\Plant<N>\Log\Rtuphase1\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean power L1 in kW	1	1	1	1	1	1
Maximum power L1 in kW	2	2	2	2	2	2
Minimum power L1 in kW	3	3	3	3	3	3
Mean power L2 in kW	4	4	4	4	4	4
Maximum power L2 in kW	5	5	5	5	5	5
Minimum power L2 in kW	6	6	6	6	6	6
Mean power L3 in kW	7	7	7	7	7	7
Maximum power L3 in kW	8	8	8	8	8	8
Minimum power L3 in kW	9	9	9	9	9	9
Mean reactive power L1 in kVAr	10	10	10	10	10	10
Maximum reactive power L1 in kVAr	11	11	11	11	11	11
Minimum reactive power L1 in kVAr	12	12	12	12	12	12
Mean reactive power L2 in kVAr	13	13	13	13	13	13
Maximum reactive power L2 in kVAr	14	14	14	14	14	14
Minimum reactive power L2 in kVAr	15	15	15	15	15	15
Mean reactive power L3 in kVAr	16	16	16	16	16	16
Maximum reactive power L3 in kVAr	17	17	17	17	17	17
Minimum reactive power L3 in kVAr	18	18	18	18	18	18

3.5.1.4 Loc\Ctrl\Plant<N>\Para node of the RTU

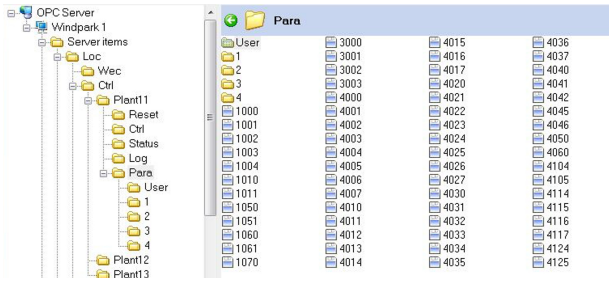


Fig. 21: Loc\Ctrl\Plant<N>\Para node of the RTU

The *Loc\Ctrl\Plant<N>\Para* node can be used to view the control parameters of an ENERCON SCADA device. The item names correspond to the parameter names in the device software.

The *Loc\Ctrl\Plant<N>\Para\User* node also includes items whose names correspond to the names of the parameters in the device software. The values of these items indicate which user changed the relevant parameter most recently.

3.5.2 FCU

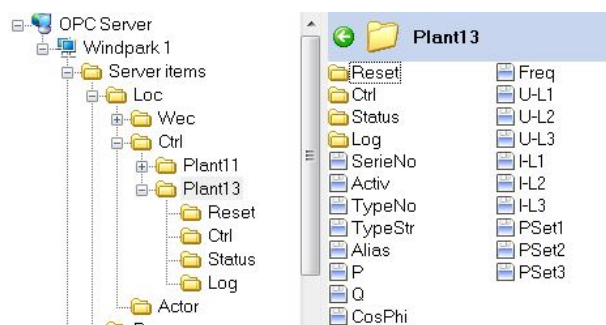


Fig. 22: Loc\Ctrl\Plant<N> node of the FCU

The nodes and items of the ENERCON SCADA system's Farm Control Unit (FCU) are described below.

Tab. 54: Loc\Ctrl\Plant<N> node of the FCU

Node/item	Description
Reset	Reset of the control system
Ctrl	Items for FCU control
Status	FCU status data
Log	FCU report data
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
P	Power in kW
Q	Reactive power in kVAr
CosPhi	Power factor Cos Phi
Freq	Frequency in Hz
U-L1	Voltage U12 in V
U-L2	Voltage U23 in V
U-L3	Voltage U31 in V
I-L1	Current L1 in A
I-L2	Current L2 in A
I-L3	Current L3 in A
PSet1	Active power setpoint 1 in kW
PSet2	Active power setpoint 2 in kW
PSet3	Active power setpoint 3 in kW

3.5.2.1 Loc\Ctrl\Plant<N>\Log\Fcuctr1\<Period> node

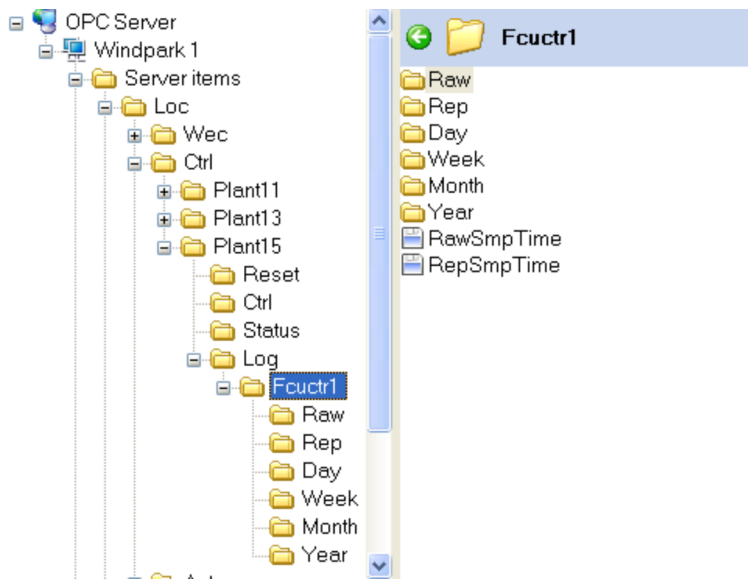


Fig. 23: Loc\Ctrl\Plant<N>\Log\Fcuctr1\<Period> node

This node contains the report data of the FCU data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 55: Loc\Ctrl\Plant<N>\Log\Fcuctr1\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean power in kW	1	1	1	1	1	1
Maximum power in kW	2	2	2	2	2	2
Minimum power in kW	3	3	3	3	3	3
Mean reactive power in kVAr	4	4	4	4	4	4
Maximum reactive power in kVAr	5	5	5	5	5	5
Minimum reactive power in kVAr	6	6	6	6	6	6
Mean Cos Phi	7	7	7	7	7	7
Maximum Cos Phi	8	8	8	8	8	8
Minimum Cos Phi	9	9	9	9	9	9
Mean frequency in Hz	10	10	10	10	10	10
Maximum frequency in Hz	11	11	11	11	11	11
Minimum frequency in Hz	12	12	12	12	12	12
Mean voltage U12 in V	13	13	13	13	13	13
Maximum voltage U12 in V	14	14	14	14	14	14
Minimum voltage U12 in V	15	15	15	15	15	15
Mean voltage U23 in V	16	16	16	16	16	16

Description	Raw	Rep	Day	Week	Month	Year
Maximum voltage U23 in V	17	17	17	17	17	17
Minimum voltage U23 in V	18	18	18	18	18	18
Mean voltage U31 in V	19	19	19	19	19	19
Maximum voltage U31 in V	20	20	20	20	20	20
Minimum voltage U31 in V	21	21	21	21	21	21
Mean current L1 in A	22	22	22	22	22	22
Maximum current L1 in A	23	23	23	23	23	23
Minimum current L1 in A	24	24	24	24	24	24
Mean current L2 in A	25	25	25	25	25	25
Maximum current L2 in A	26	26	26	26	26	26
Minimum current L2 in A	27	27	27	27	27	27
Mean current L3 in A	28	28	28	28	28	28
Maximum current L3 in A	29	29	29	29	29	29
Minimum current L3 in A	30	30	30	30	30	30
Mean active power setpoint 1 in kW	31	31	31	31	31	31
Mean active power setpoint 2 in kW	32	32	32	32	32	32
Mean active power setpoint 3 in kW	33	33	33	33	33	33

3.5.3 FCU E2

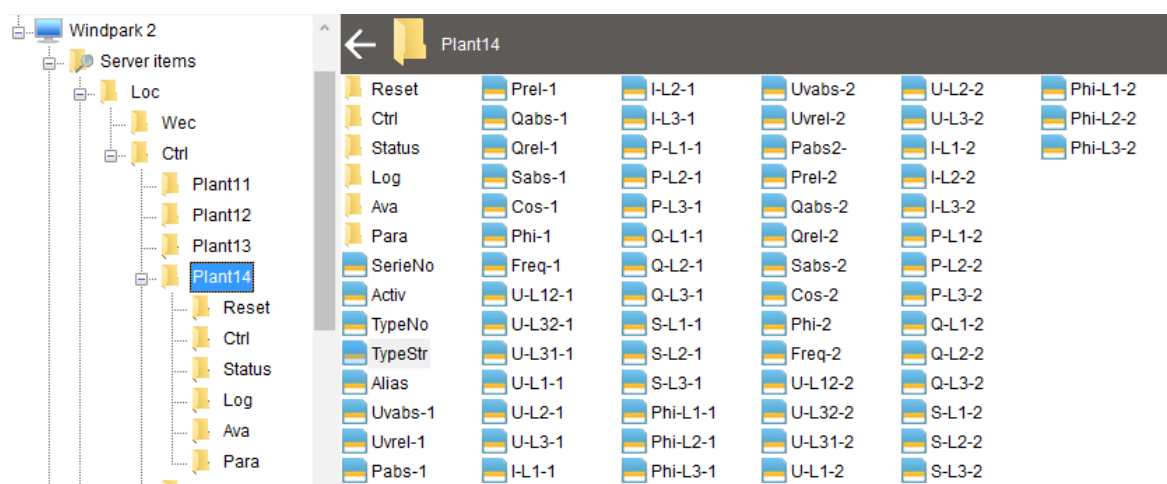


Fig. 24: Loc\Ctrl\Plant<N> node of the FCU E2

The nodes and items of the ENERCON SCADA system's Farm Control Unit E2 (FCU E2) are described below.

Tab. 56: Loc\Ctrl\Plant<N> node of the FCU E2

Node/item	Description
Reset	Reset of the control system
Ctrl	Items for controlling the FCU E2
Status	FCU E2 status data
Log	FCU E2 report data
Ava	Availability data of the FCU E2
Para	Parameters of the FCU E2
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
Uvabs-1/2	Voltage in V
Uvrel-1/2	Voltage in %
Pabs-1/2	Active power in kW
Prel-1/2	Active power in %
Qabs-1/2	Reactive power in kVAr
Qrel-1/2	Reactive power in %
Sabs-1/2	Apparent power in VA
Cos-1/2	Power factor
Phi-1/2	Phase angle in °
Freq-1/2	Frequency in Hz
U-L12-1/2	Voltage U12 in V (measurement 1/2)

Node/item	Description
U-L32-1/2	Voltage U32 in V (measurement 1/2)
U-L31-1/2	Voltage U31 in V (measurement 1/2)
U-L1-1/2	Voltage U1 in V (measurement 1/2)
U-L2-1/2	Voltage U2 in V (measurement 1/2)
U-L3-1/2	Voltage U3 in V (measurement 1/2)
I-L1-1/2	Current L1 in A (measurement 1/2)
I-L2-1/2	Current L2 in A (measurement 1/2)
I-L3-1/2	Current L3 in A (measurement 1/2)
P-L1-1	Active power L1 in kW
P-L2-1	Active power L2 in kW
P-L3-1	Active power L3 in kW
Q-L1-1/2	Reactive power L1 in kVAr
Q-L2-1/2	Reactive power L2 in kVAr
Q-L3-1/2	Reactive power L3 in kVAr
S-L1-1/2	Apparent power L1 in VA
S-L2-1/2	Apparent power L2 in VA
S-L3-1/2	Apparent power L3 in VA
Phi-L1-1/2	Phase angle L1 in °
Phi-L2-1/2	Phase angle L2 in °
Phi-L3-1/2	Phase angle L3 in °

3.5.3.1 Loc\Ctr\Plant<N>\Log\Fcu2ctr1<Period> node

The node contains the report data from the 1st data logger (measurement 1) of the FCU E2 over different periods of time. The report data is also applicable for the 2nd data logger (measurement 2) of the FCU E2 (Loc\Ctr\Plant<N>\Log\Fcu2ctr2<Period>).

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 57: Loc\Ctr\Plant<N>\Log\Fcu2ctr1<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean voltage in V	1	1	1	1	1	1
Maximum voltage in V	2	2	2	2	2	2
Minimum voltage in V	3	3	3	3	3	3
Mean power in kW	4	4	4	4	4	4
Maximum power in kW	5	5	5	5	5	5

Description	Raw	Rep	Day	Week	Month	Year
Minimum power in kW	6	6	6	6	6	6
Mean reactive power in kVAr	7	7	7	7	7	7
Maximum reactive power in kVAr	8	8	8	8	8	8
Minimum reactive power in kVAr	9	9	9	9	9	9
Mean apparent power in VA	10	10	10	10	10	10
Maximum apparent power in VA	11	11	11	11	11	11
Minimum apparent power in VA	12	12	12	12	12	12
Mean Phi in °	13	13	13	13	13	13
Maximum Phi in °	14	14	14	14	14	14
Minimum Phi in °	15	15	15	15	15	15
Mean frequency in Hz	16	16	16	16	16	16
Maximum frequency in Hz	17	17	17	17	17	17
Minimum frequency in Hz	18	18	18	18	18	18
Mean voltage U12 in V	19	19	19	19	19	19
Maximum voltage U12 in V	20	20	20	20	20	20
Minimum voltage U12 in V	21	21	21	21	21	21
Mean voltage U23 in V	22	22	22	22	22	22
Maximum voltage U23 in V	23	23	23	23	23	23
Minimum voltage U23 in V	24	24	24	24	24	24
Mean voltage U21 in V	25	25	25	25	25	25
Maximum voltage U31 in V	26	26	26	26	26	26
Minimum voltage U31 in V	27	27	27	27	27	27
Mean voltage U1 in V	28	28	28	28	28	28
Maximum voltage U1 in V	29	29	29	29	29	29
Minimum voltage U1 in V	30	30	30	30	30	30
Mean voltage U2 in V	31	31	31	31	31	31
Maximum voltage U2 in V	32	32	32	32	32	32
Minimum voltage U2 in V	33	33	33	33	33	33
Mean voltage U3 in V	34	34	34	34	34	34
Maximum voltage U3 in V	35	35	35	35	35	35
Minimum voltage U3 in V	36	36	36	36	36	36
Mean current L1 in A	37	37	37	37	37	37
Maximum current L1 in A	38	38	38	38	38	38
Minimum current L1 in A	39	39	39	39	39	39
Mean current L2 in A	40	40	40	40	40	40
Maximum current L2 in A	41	41	41	41	41	41

Description	Raw	Rep	Day	Week	Month	Year
Minimum current L2 in A	42	42	42	42	42	42
Mean current L3 in A	43	43	43	43	43	43
Maximum current L3 in A	44	44	44	44	44	44
Minimum current L3 in A	45	45	45	45	45	45
Mean power L1 in kW	46	46	46	46	46	46
Maximum power L1 in kW	47	47	47	47	47	47
Minimum power L1 in kW	48	48	48	48	48	48
Mean power L2 in kW	49	49	49	49	49	49
Maximum power L2 in kW	50	50	50	50	50	50
Minimum power L2 in kW	51	51	51	51	51	51
Mean power L3 in kW	52	52	52	52	52	52
Maximum power L3 in kW	53	53	53	53	53	53
Minimum power L3 in kW	54	54	54	54	54	54
Mean apparent power L1 in VA	55	55	55	55	55	55
Maximum apparent power L1 in VA	56	56	56	56	56	56
Minimum apparent power L1 in VA	57	57	57	57	57	57
Mean apparent power L2 in VA	58	58	58	58	58	58
Maximum apparent power L2 in VA	59	59	59	59	59	59
Minimum apparent power L2 in VA	60	60	60	60	60	60
Mean apparent power L3 in VA	61	61	61	61	61	61
Maximum apparent power L3 in VA	62	62	62	62	62	62
Minimum apparent power L3 in VA	63	63	63	63	63	63
Mean reactive power L1 in kVAr	64	64	64	64	64	64
Maximum reactive power L1 in kVAr	65	65	65	65	65	65
Minimum reactive power L1 in kVAr	66	66	66	66	66	66
Mean reactive power L2 in kVAr	67	67	67	67	67	67
Maximum reactive power L2 in kVAr	68	68	68	68	68	68
Minimum reactive power L2 in kVAr	69	69	69	69	69	69
Mean reactive power L3 in kVAr	70	70	70	70	70	70
Maximum reactive power L3 in kVAr	71	71	71	71	71	71
Minimum reactive power L3 in kVAr	72	72	72	72	72	72
Mean phase angle L1 in °	73	73	73	73	73	73
Maximum phase angle L1 in °	74	74	74	74	74	74
Minimum phase angle L1 in °	75	75	75	75	75	75
Mean phase angle L2 in °	76	76	76	76	76	76
Maximum phase angle L2 in °	77	77	77	77	77	77

Description	Raw	Rep	Day	Week	Month	Year
Minimum phase angle L2 in °	78	78	78	78	78	78
Mean phase angle L3 in °	79	79	79	79	79	79
Maximum phase angle L3 in °	80	80	80	80	80	80
Minimum phase angle L3 in °	81	81	81	81	81	81
Energy meter, production in kWh	82	82	82	82	82	82
Energy meter, consumption in kWh	83	83	83	83	83	83
Reactive power meter (inductive) in kVAR	84	84	84	84	84	84
Reactive power meter (capacitive) in kVAR	85	85	85	85	85	85
Operating hours in h	86	86	86	86	86	86

3.5.4 GDA

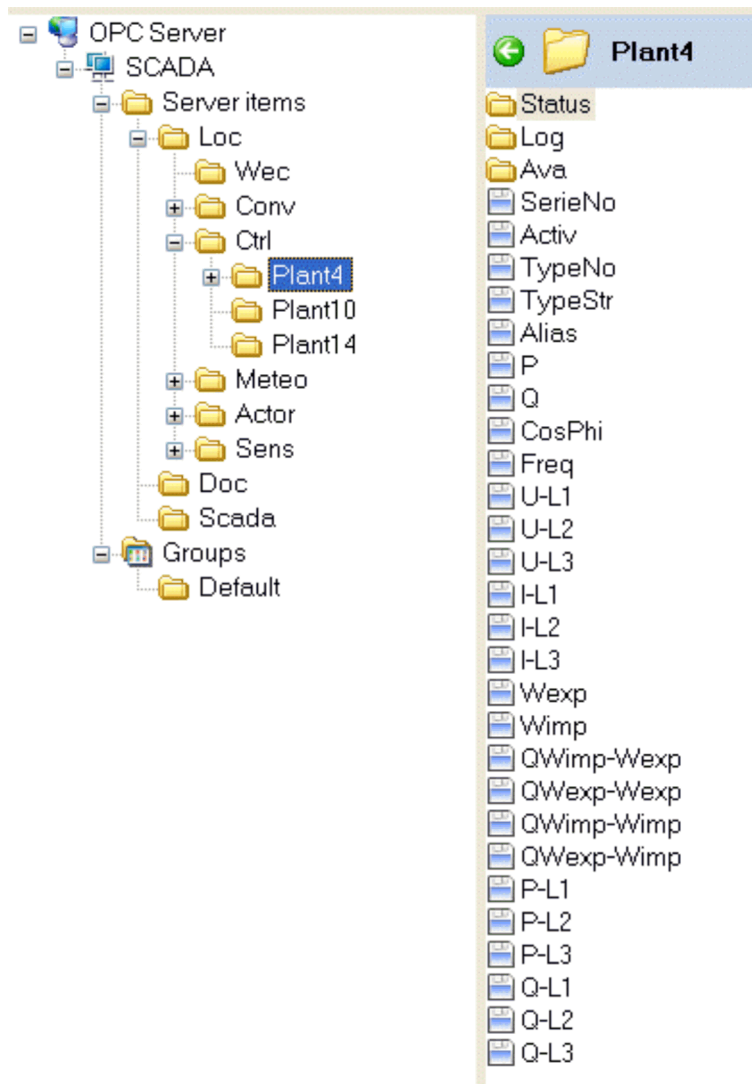


Fig. 25: Loc/Ctrl/Plant<N> node of the GDA

The nodes and items of the ENERCON SCADA GDA are described below.

Tab. 58: Loc/Ctrl/Plant<N> node of the GDA

Node/item	Description
Status	GDA status data
Log	GDA report data
Ava	GDA availability data
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
P	Power in kW
Q	Reactive power in kVAr

Node/item	Description
CosPhi	Power factor Cos Phi
Freq	Frequency in Hz
U-L1	Low voltage U12 in V
U-L2	Low voltage U23 in V
U-L3	Low voltage U31 in V
I-L1	Power feed L1 in A
I-L2	Power feed L2 in A
I-L3	Power feed L3 in A
Wexp	Exported energy in kWh
Wimp	Imported energy in kWh
QWimp-Wexp	Reactive energy import during active energy export in kVArh
QWexp-Wexp	Reactive energy export during active energy export in kVArh
QWimp-Wimp	Reactive energy import during active energy import in kVArh
QWexp-Wimp	Reactive energy export during active energy import in kVArh
P-L1	Power L1 in kW
P-L2	Power L2 in kW
P-L3	Power L3 in kW
Q-L1	Reactive power L1 in kVAr
Q-L2	Reactive power L2 in kVAr
Q-L3	Reactive power L3 in kVAr

3.5.4.1 Loc\Ctrl\Plant<N>\Log\Farmctr1\<Period> node

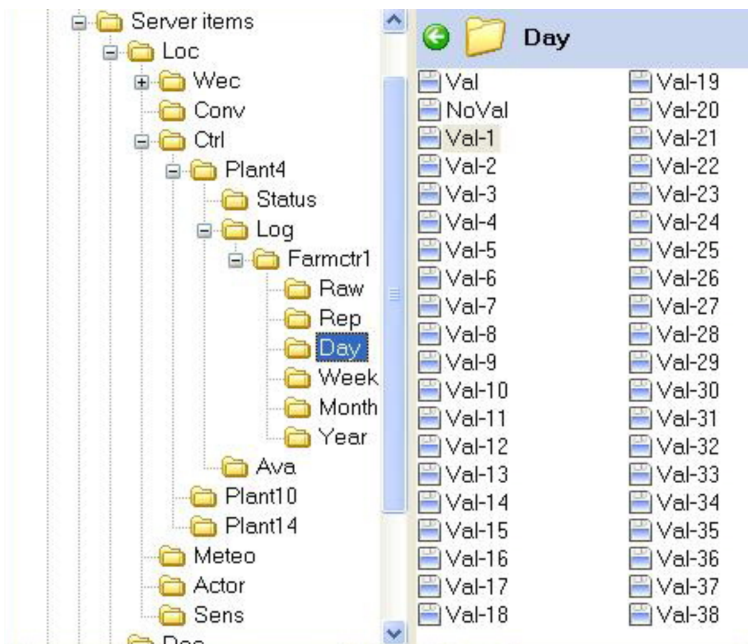


Fig. 26: Loc\Ctrl\Plant<N>\Log\Farmctr1\Day node

This node contains the report data of the GDA data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val*-<X> are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 59: Loc\Ctrl\Plant<N>\Log\Farmctr1\Day node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean power in kW	1	1	1	1	1	1
Maximum power in kW	2	2	2	2	2	2
Minimum power in kW	3	3	3	3	3	3
Mean reactive power in kVAr	4	4	4	4	4	4
Maximum reactive power in kVAr	5	5	5	5	5	5
Minimum reactive power in kVAr	6	6	6	6	6	6
Mean Cos Phi	7	7	7	7	7	7
Maximum Cos Phi	8	8	8	8	8	8
Minimum Cos Phi	9	9	9	9	9	9
Mean frequency in Hz	10	10	10	10	10	10
Maximum frequency in Hz	11	11	11	11	11	11
Minimum frequency in Hz	12	12	12	12	12	12
Mean voltage U12 in V	13	13	13	13	13	13
Maximum voltage U12 in V	14	14	14	14	14	14

Description	Raw	Rep	Day	Week	Month	Year
Minimum voltage U12 in V	15	15	15	15	15	15
Mean voltage U23 in V	16	16	16	16	16	16
Maximum voltage U23 in V	17	17	17	17	17	17
Minimum voltage U23 in V	18	18	18	18	18	18
Mean voltage U31 in V	19	19	19	19	19	19
Maximum voltage U31 in V	20	20	20	20	20	20
Minimum voltage U31 in V	21	21	21	21	21	21
Mean current L1 in A	22	22	22	22	22	22
Maximum current L1 in A	23	23	23	23	23	23
Minimum current L1 in A	24	24	24	24	24	24
Mean current L2 in A	25	25	25	25	25	25
Maximum current L2 in A	26	26	26	26	26	26
Minimum current L2 in A	27	27	27	27	27	27
Mean current L3 in A	28	28	28	28	28	28
Maximum current L3 in A	29	29	29	29	29	29
Minimum current L3 in A	30	30	30	30	30	30
Exported energy in kWh	31	31	31	31	31	31
Imported energy in kWh	32	32	32	32	32	32
Reactive power (inductive) 1 in kVAr	33	33	33	33	33	33
Reactive power (capacitive) 1 in kVAr	34	34	34	34	34	34
Reactive power (inductive) 2 in kVAr	35	35	35	35	35	35
Reactive power (capacitive) 2 in kVAr	36	36	36	36	36	36

3.5.4.2 Loc\Ctrl\Plant<N>\Log\Farmctr15<Period> node (optional)

This optional node contains the report data of the ENERCON GDA data logger as 15-minute data for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 60: Loc\Ctrl\Plant<N>\log\Farmctr15<Period> node (optional)

Description	Raw	Rep
Number of sampling values	0	0
Mean power in kW	1	1
Mean reactive power in kVAr	2	2
Energy meter, production in kWh	3	3
Energy meter, consumption in kWh	4	4
Inductive reactive power meter during active power generation in kVAr	5	5
Capacitive reactive power meter during active power generation in kVAr	6	6
Inductive reactive power meter during active power consumption in kVAr	7	7
Capacitive reactive power meter during active power consumption in kVAr	8	8

3.5.5 SCU

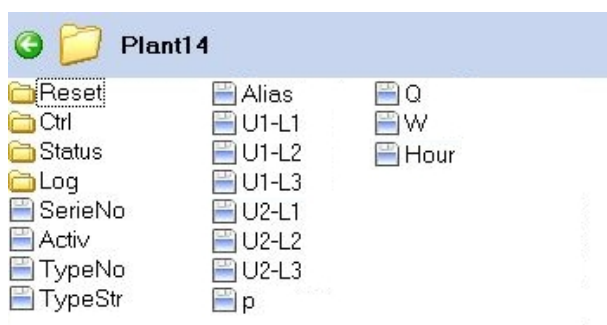


Fig. 27: Loc\Ctrl\Plant<N> node of the SCU

The nodes and items of the ENERCON SCADA SCU are described below.

Tab. 61: Loc\Ctrl\Plant<N> node of the SCU

Node/item	Description
Reset	Reset of the control system
Status	SCU status data
Log	SCU report data
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
U1-L1	Voltage level 1 U12 in V
U1-L2	Voltage level 1 U23 in V
U1-L3	Voltage level 1 U31 in V
U2-L1	Voltage level 2 U12 in V
U2-L2	Voltage level 2 U23 in V
U2-L3	Voltage level 2 U31 in V
P	Power in kW
Q	Reactive power in kVAr
W	Energy in kWh
Hour	Operating hours in h

3.5.5.1 Loc\Ctrl\Plant<N>\Log\Substat1\<Period> node

This node contains the report data of the SCU data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 62: Loc\Ctrl\Plant<N>\Log\Substat1\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean voltage 1 U12 in V	1	1	1	1	1	1
Mean voltage 1 U23 in V	2	2	2	2	2	2
Mean voltage 1 U31 in V	3	3	3	3	3	3
Mean voltage 2 U12 in V	4	4	4	4	4	4
Mean voltage 2 U23 in V	5	5	5	5	5	5
Mean voltage 2 U31 in V	6	6	6	6	6	6
Mean power in kW	7	7	7	7	7	7
Mean reactive power in kVAr	8	8	8	8	8	8

3.5.6 Transmission substation 2

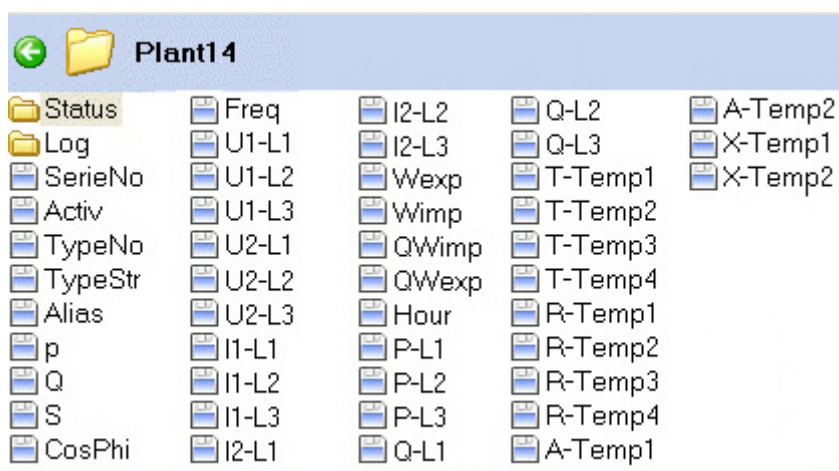


Fig. 28: The Loc/Ctrl/Plant<N> node for transmission substation 2

The nodes and items of the devices with control system type *Transmission substation 2* are described below.

Tab. 63: Loc/Ctrl/Plant<N> node for transmission substation 2

Node/item	Description
Status	Device status data
Log	Device report data
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
P	Power in kW
Q	Reactive power in kVAr
S	Apparent power in VA
CosPhi	Power factor Cos Phi
Freq	Frequency in Hz
U1-L1	Voltage level 1 U12 in V
U1-L2	Voltage level 1 U23 in V
U1-L3	Voltage level 1 U31 in V
U2-L1	Voltage level 2 U12 in V
U2-L2	Voltage level 2 U23 in V
U2-L3	Voltage level 2 U31 in V
I1-L1	Current level 1 L1 in A
I1-L2	Current level 1 L2 in A
I1-L3	Current level 1 L3 in A
I2-L1	Current level 2 L1 in A

Node/item	Description
I2-L2	Current level 2 L2 in A
I2-L3	Current level 2 L3 in A
Wexp	Exported energy in kWh
Wimp	Imported energy in kWh
QWimp	Reactive power import in kVARh
QWexp	Reactive power export in kVARh
Hour	Operating hours in h
P-L1	Power L1 in kWh
P-L2	Power L2 in kWh
P-L3	Power L3 in kWh
Q-L1	Reactive power L1 in kVAR
Q-L2	Reactive power L2 in kVAR
Q-L3	Reactive power L3 in kVAR
T-Temp1 ¹	Transformer temperature 1 in °C
T-Temp2 ¹	Transformer temperature 2 in °C
T-Temp3 ¹	Transformer temperature 3 in °C
T-Temp4 ¹	Transformer temperature 4 in °C
R-Temp1 ¹	Room temperature 1 in °C
R-Temp2 ¹	Room temperature 2 in °C
R-Temp3 ¹	Room temperature 3 in °C
R-Temp4 ¹	Room temperature 4 in °C
A-Temp1 ¹	Ambient temperature 1 in °C
A-Temp2 ¹	Ambient temperature 2 in °C
X-Temp1 ¹	Temperature in °C (dummy)
X-Temp2 ¹	Temperature in °C (dummy)

¹ The positions of the temperature sensors are defined for each specific project.

3.5.6.1 Loc\Ctrl\Plant<N>\Log\Substat2\<Period> node

This node contains the report data of the data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 64: Loc\Ctrl\Plant<N>\Log\Substat2\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean power in kW	1	1	1	1	1	1
Maximum power in kW	2	2	2	2	2	2
Minimum power in kW	3	3	3	3	3	3
Mean reactive power in kVAr	4	4	4	4	4	4
Maximum reactive power in kVAr	5	5	5	5	5	5
Minimum reactive power in kVAr	6	6	6	6	6	6
Mean apparent power in kVA	7	7	7	7	7	7
Maximum apparent power in kVA	8	8	8	8	8	8
Minimum apparent power in kVA	9	9	9	9	9	9
Mean Cos Phi	10	10	10	10	10	10
Maximum Cos Phi	11	11	11	11	11	11
Minimum Cos Phi	12	12	12	12	12	12
Mean frequency in Hz	13	13	13	13	13	13
Maximum frequency in Hz	14	14	14	14	14	14
Minimum frequency in Hz	15	15	15	15	15	15
Mean voltage 1 L1 in V	16	16	16	16	16	16
Maximum voltage 1 L1 in V	17	17	17	17	17	17
Minimum voltage 1 L1 in V	18	18	18	18	18	18
Mean voltage 1 L2 in V	19	19	19	19	19	19
Maximum voltage 1 L2 in V	20	20	20	20	20	20
Minimum voltage 1 L2 in V	21	21	21	21	21	21
Mean voltage 1 L3 in V	22	22	22	22	22	22
Maximum voltage 1 L3 in V	23	23	23	23	23	23
Minimum voltage 1 L3 in V	24	24	24	24	24	24
Mean voltage 2 L1 in V	25	25	25	25	25	25
Maximum voltage 2 L1 in V	26	26	26	26	26	26
Minimum voltage 2 L1 in V	27	27	27	27	27	27
Mean voltage 2 L2 in V	28	28	28	28	28	28

Description	Raw	Rep	Day	Week	Month	Year
Maximum voltage 2 L2 in V	29	29	29	29	29	29
Minimum voltage 2 L2 in V	30	30	30	30	30	30
Mean voltage 2 L3 in V	31	31	31	31	31	31
Maximum voltage 2 L3 in V	32	32	32	32	32	32
Minimum voltage 2 L3 in V	33	33	33	33	33	33
Mean current 1 L1 in A	34	34	34	34	34	34
Maximum current 1 L1 in A	35	35	35	35	35	35
Minimum current 1 L1 in A	36	36	36	36	36	36
Mean current 1 L2 in A	37	37	37	37	37	37
Maximum current 1 L2 in A	38	38	38	38	38	38
Minimum current 1 L2 in A	39	39	39	39	39	39
Mean current 1 L3 in A	40	40	40	40	40	40
Maximum current 1 L3 in A	41	41	41	41	41	41
Minimum current 1 L3 in A	42	42	42	42	42	42
Mean current 2 L1 in A	43	43	43	43	43	43
Maximum current 2 L1 in A	44	44	44	44	44	44
Minimum current 2 L1 in A	45	45	45	45	45	45
Mean current 2 L2 in A	46	46	46	46	46	46
Maximum current 2 L2 in A	47	47	47	47	47	47
Minimum current 2 L2 in A	48	48	48	48	48	48
Mean current 2 L3 in A	49	49	49	49	49	49
Maximum current 2 L3 in A	50	50	50	50	50	50
Minimum current 2 L3 in A	51	51	51	51	51	51
Exported energy in kWh	52	52	52	52	52	52
Imported energy in kWh	53	53	53	53	53	53
Reactive power (inductive) in kVAr	54	54	54	54	54	54
Reactive power (capacitive) in kVAr	55	55	55	55	55	55
Operating hours in h	56	56	56	56	56	56

3.5.6.2 Loc\Ctrl\Plant<N>\Log\SubTemp2\<Period> node

The node contains the report data of the data logger for various periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 65: Loc\Ctrl\Plant<N>\Log\SubTemp2\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean temperature 1 in °C	1	1	1	1	1	1
Maximum temperature 1 in °C	2	2	2	2	2	2
Minimum temperature 1 in °C	3	3	3	3	3	3
Mean temperature 2 in °C	4	4	4	4	4	4
Maximum temperature 2 in °C	5	5	5	5	5	5
Minimum temperature 2 in °C	6	6	6	6	6	6
Mean temperature 3 in °C	7	7	7	7	7	7
Maximum temperature 3 in °C	8	8	8	8	8	8
Minimum temperature 3 in °C	9	9	9	9	9	9
Mean temperature 4 in °C	10	10	10	10	10	10
Maximum temperature 4 in °C	11	11	11	11	11	11
Minimum temperature 4 in °C	12	12	12	12	12	12
Mean temperature 5 in °C	13	13	13	13	13	13
Maximum temperature 5 in °C	14	14	14	14	14	14
Minimum temperature 5 in °C	15	15	15	15	15	15
Mean temperature 6 in °C	16	16	16	16	16	16
Maximum temperature 6 in °C	17	17	17	17	17	17
Minimum temperature 6 in °C	18	18	18	18	18	18
Mean temperature 7 in °C	19	19	19	19	19	19
Maximum temperature 7 in °C	20	20	20	20	20	20
Minimum temperature 7 in °C	21	21	21	21	21	21
Mean temperature 8 in °C	22	22	22	22	22	22
Maximum temperature 8 in °C	23	23	23	23	23	23
Minimum temperature 8 in °C	24	24	24	24	24	24
Mean temperature 9 in °C	25	25	25	25	25	25
Maximum temperature 9 in °C	26	26	26	26	26	26
Minimum temperature 9 in °C	27	27	27	27	27	27
Mean temperature 10 in °C	28	28	28	28	28	28

Description	Raw	Rep	Day	Week	Month	Year
Maximum temperature 10 in °C	29	29	29	29	29	29
Minimum temperature 10 in °C	30	30	30	30	30	30
Mean temperature 11 in °C	31	31	31	31	31	31
Maximum temperature 11 in °C	32	32	32	32	32	32
Minimum temperature 11 in °C	33	33	33	33	33	33
Mean temperature 12 in °C	34	34	34	34	34	34
Maximum temperature 12 in °C	35	35	35	35	35	35
Minimum temperature 12 in °C	36	36	36	36	36	36

3.5.6.3 Loc\Ctrl\Plant<N>\Log\Subphase2\<Period> node

This node contains the report data of the data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 66: Loc\Ctrl\Plant<N>\Log\Subphase2\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean power L1 in kW	1	1	1	1	1	1
Maximum power L1 in kW	2	2	2	2	2	2
Minimum power L1 in kW	3	3	3	3	3	3
Mean power L2 in kW	4	4	4	4	4	4
Maximum power L2 in kW	5	5	5	5	5	5
Minimum power L2 in kW	6	6	6	6	6	6
Mean power L3 in kW	7	7	7	7	7	7
Maximum power L3 in kW	8	8	8	8	8	8
Minimum power L3 in kW	9	9	9	9	9	9
Mean reactive power L1 in kVAr	10	10	10	10	10	10
Maximum reactive power L1 in kVAr	11	11	11	11	11	11
Minimum reactive power L1 in kVAr	12	12	12	12	12	12
Mean reactive power L2 in kVAr	13	13	13	13	13	13
Maximum reactive power L2 in kVAr	14	14	14	14	14	14
Minimum reactive power L2 in kVAr	15	15	15	15	15	15
Mean reactive power L3 in kVAr	16	16	16	16	16	16
Maximum reactive power L3 in kVAr	17	17	17	17	17	17
Minimum reactive power L3 in kVAr	18	18	18	18	18	18

3.6 Loc\Meteo node

The node *Meteo* contains the data of the *meteorological devices* group.

3.6.1 Loc\Meteo\Plant<N> node

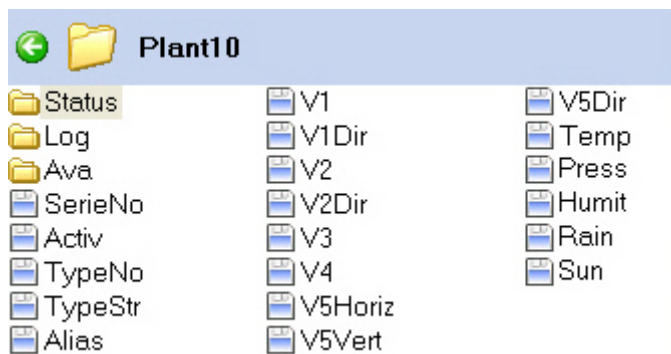


Fig. 29: Loc\Meteo\Plant<N> node

The node *Plant<N>* contains the ENERCON SCADA METEO data.

Tab. 67: Loc\Meteo\Plant<N> node

Node/Item	Description
Status	METEO status data
Log	METEO report data
Ava	METEO availability data
SerieNo	Serial number assigned by ENERCON
Activ	Plant is either in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer
V1	Wind speed at height 1 in m/s
V1Dir	Wind direction at height 1 in °
V2	Wind speed at height 2 in m/s
V2Dir	Wind direction at height 2 in °
V3	Wind speed at height 3 in m/s
V4	Wind speed at height 4 in m/s
V5Horiz	Wind speed (horizontal) at height 5 in m/s
V5Vert	Wind speed (vertical) at height 5 in m/s
V5Dir	Wind direction at height 5 in °
Temp	Outside temperature in °C
Press	Air pressure in mbar
Humit	Air humidity in %
Rain	Precipitation in mm/min
Sun	Solar radiation in W/m ²

3.6.2 Loc\Meteo\Plant<N>\Log\Wind1\<Period> node

The node <Period> contains the report data of the ENERCON SCADA METEO data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val-<X>* are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 68: Loc\Meteo\Plant<N>\Log\Wind1\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Mean wind speed 1 in m/s	1	1	1	1	1	1
Maximum wind speed 1 in m/s	2	2	2	2	2	2
Minimum wind speed 1 in m/s	3	3	3			
Wind direction 1 in °	4	4	4			
Mean wind speed 2 in m/s	5	5	5	3	3	3
Maximum wind speed 2 in m/s	6	6	6	4	4	4
Minimum wind speed 2 in m/s	7	7	7			
Wind direction 2 in °	8	8	8			
Mean X wind direction 1 in °	9	9	9			
Mean Y wind direction 1 in °	10	10	10			
Mean X wind direction 2 in °	11	11	11			
Mean Y wind direction 2 in °	12	12	12			
Mean wind speed 3 in m/s	13	13	13	5	5	5
Maximum wind speed 3 in m/s	14	14	14	6	6	6
Minimum wind speed 3 in m/s	15	15	15			
Mean wind speed 4 in m/s	16	16	16	7	7	7
Maximum wind speed 4 in m/s	17	17	17	8	8	8
Minimum wind speed 4 in m/s	18	18	18			
Mean horizontal wind speed in m/s ¹	19	19	19	9	9	9
Maximum horizontal wind speed in m/s ¹	20	20	20	10	10	10
Minimum horizontal wind speed in m/s ¹	21	21	21			
Mean vertical wind speed in m/s ¹	22	22	22	11	11	11
Maximum vertical wind speed in m/s ¹	23	23	23	12	12	12
Minimum vertical wind speed in m/s ¹	24	24	24			
Mean X wind direction in ° ¹	25	25	25			
Mean Y wind direction in ° ¹	26	26	26			

¹ Only available if ultrasonic anemometer is installed

3.6.3 Loc\Meteo\Plant<Index>\Log\Weather2\<Period> node

The node <Period> contains the report data of the ENERCON SCADA METEO weather-measurement data logger for different periods of time.

The *NoVal* item shows the number of written data records.

The data records *Val*-<X> are arrays and indicate the data described in the table. The actual scope of the data can vary depending on the time period (*Raw*, *Rep*, etc.). The number in the table indicates the position (index) of the data value within the array. If no number is specified, this means that the data value is not available for this time period.

Tab. 69: Loc\Meteo\Plant<Index>\Log\Weather2\<Period> node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Ambient temperature in °C	1	1	1	1	1	1
Atmospheric pressure in mbar	2	2	2	2	2	2
Air humidity in %	3	3	3	3	3	3
Precipitation in mm/min	4	4	4	4	4	4
Solar radiation in W/m ²	5	5	5	5	5	5

3.7 Loc\Actor node

This node contains the data of the *actors* group.

3.7.1 PDI

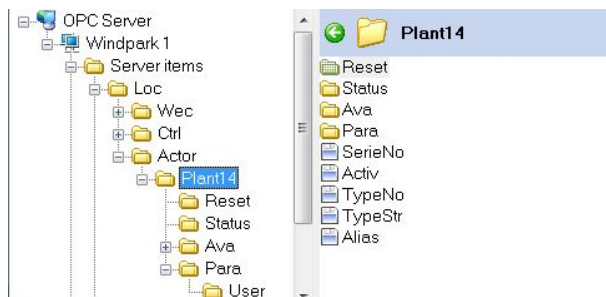


Fig. 30: Loc/Actor/Plant<N> node of the PDI

The nodes and items of an ENERCON SCADA PDI are described below.

Tab. 70: Loc/Actor/Plant<N> node of the PDI

Node/item	Description
Reset	Reset of the control system
Status	PDI status data
Log	PDI data loggers
Ava	PDI availability data
Para	PDI parameters
SerieNo	Serial number assigned by ENERCON
Activ	Plant is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	ID that can be selected by the customer

3.7.1.1 Loc\Actor\Plant<N>\Para node of the PDI

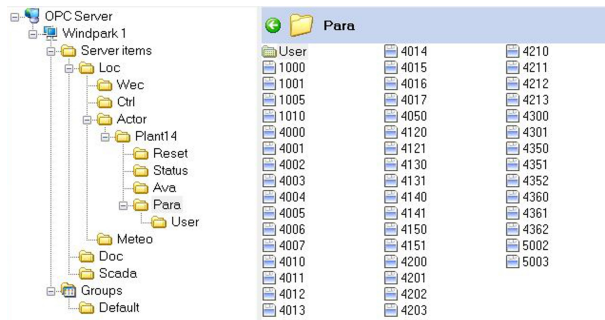


Fig. 31: Loc\Actor\Plant<N>\Para node of the PDI

The *Para* node can be used to view the control parameters of an ENERCON SCADA device. The item names correspond to the parameter names in the device software.

The *Loc\Actor\Plant<N>\Para\User* node also includes items whose names correspond to the names of the parameters in the device software. The values of these items indicate which user changed the relevant parameter most recently.

3.8 The Doc (documentation) node

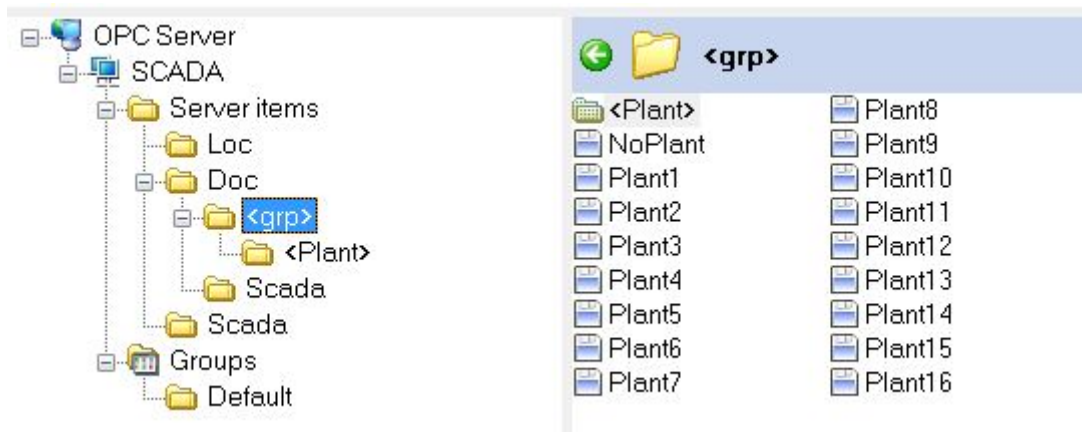


Fig. 32: The Doc (documentation) node

The items in the *Doc* node provide explanations of the corresponding items in the *Loc* node. This can also be used to look up abbreviations used for nodes and items in the ENERCON SCADA PDI-OPC.

The structure of the *Doc* node is essentially identical to that of the *Loc* node. The *<grp>* (group) node comprises all wind energy converters and ENERCON SCADA system devices for one wind farm. The total number is reflected in the *NoPlant* item.

3.9 The SCADA node

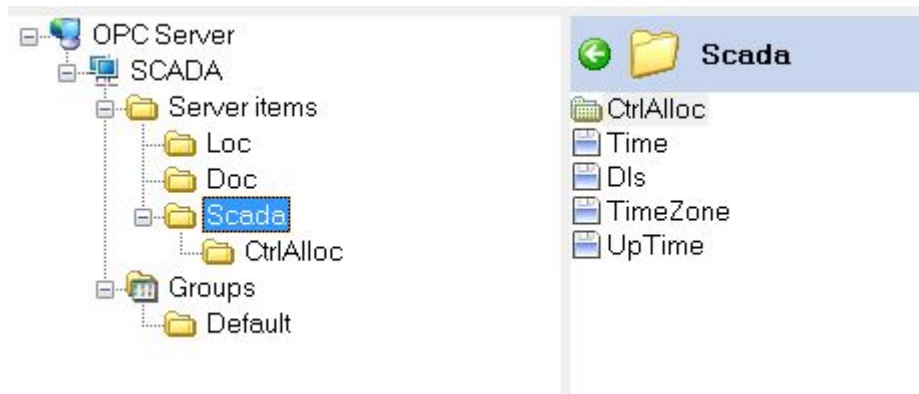


Fig. 33: The *Scada* node

The *Scada* node contains SCADA-specific items resulting from the project-specific functions of the ENERCON SCADA system.

Tab. 71: The *Scada* node

Node/item	Description
CtrlAlloc (example)	Items for global allocation of all access for control purposes
Time	Depending on the configuration, the following times are transmitted: ■ UTC activated The value transmitted is the coordinated universal time (UTC; format dd.mm.yyyy hh.mm.ss). ■ UTC deactivated The value transmitted is the local server time (format dd.mm.yyyy hh.mm.ss).
DIs	Flag for daylight saving time
TimeZone	Time zone east of the prime meridian in minutes (German time zone = +60)
UpTime	Start-up time of the OPC server, e.g. after a grid fault. This point in time is not identical to the time of installation.

4 Nodes and items for closed-loop and open-loop control

In OPC, signs (+ and -) are used in accordance with IEC 61400-21.

4.1 Session items for access for control purposes

All control interventions in the ENERCON PDI-OPC follow a uniform mechanism. For all control interventions, the session items below are therefore present in the data tree of the respective control value.

Tab. 72: Session items for control interventions

Item	Description
Session state	Current status of the session (short integer/read access). 0: Session free 1: Session reserved 2: Session parameter input 3: Wait time in loop mode 4: Session end wait time 5: Session blocked 108: Busy 109: Access denied 121: Illegal value 174: Wrong user ID 175: Insufficient access rights
Session timeout	Shows the remaining time of a session timeout (integer/read access)
2. Public key	Key generated by  server for access to that particular session (long word/read access)
1. SessionRequest 	Array for requesting a session reservation (array of 3 long words/read access). The array contains the following values in the specified sequence: Session ID Random value from the client that the client can use to determine whether a reservation was successful. This value can be read by the clients. User ID The user ID represents a user in the SCADA system. SCADA users and their rights are created in the SCADA system by ENERCON (Service IT). This value cannot be read and is therefore invisible to other clients. Private key The private key is a random value and a subkey for access to the respective session. This value cannot be read and is therefore invisible to other clients.
4. SessionSubmit 	Array for executing a session (array of 2 long words/read + write access). The array contains the following values in the specified sequence:

Item	Description
	Private key The private key registered with the server when logging on to the session. This value cannot be read and is therefore invisible to other clients.
	Public key The public key made available by the server after logging on to the session.

4.1.1 Session items for access to individual WECs for control purposes

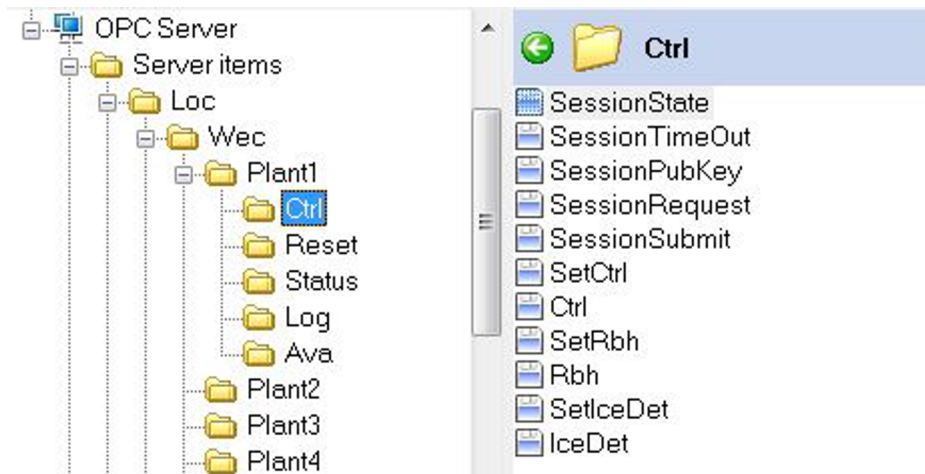


Fig. 34: Loc\Wec\Plant<N>\Ctrl node

The Loc\Wec\Plant<N>\Ctrl node contains items for access to individual wind energy converters for control interventions.

Transmission takes place in a single session.

Tab. 73: Session item for access to individual wind energy converters for control interventions

Item	Description
SetCtrl	Array for setting a control value (array of 3 long words/read + write access). The array contains the following values in the specified sequence: Control value SetCtrl [0] 0 = Start wind energy converter 1 = Stop wind energy converter at 60° blade angle 2 = Stop wind energy converter at 90° blade angle 3 = Gradient stop of wind energy converter at 60° blade angle 4 = Gradient stop of wind energy converter at 90° blade angle 5 = Ice-detection system stop 6 = Shadow-flickering stop 7 = Animal-protection stop at 60° blade angle 8 = Animal-protection stop at 90° blade angle Private key SetCtrl [1] The private key selected by the user at the start of the session. Public key SetCtrl [2] The public key made available by the server at the beginning of the session.
Ctrl	The current status of the wind energy converter (short integer/read access).

Item	Description										
	0 = WEC in operation 1 = WEC stopped at 60° blade angle 2 = WEC stopped at 90° blade angle 129 = WEC stopped at 60° blade angle with higher priority. The WEC cannot be started. 130 = WEC stopped at 90° blade angle with higher priority. The WEC cannot be started. 255: Value has not yet been initialised. Communication fault with WEC. Only operating states with values 0–2 can be changed.										
SetRbH	Array for setting the control value of the blade heating system (array of 3 long words/read + write access). The array contains the following values in the specified sequence: Control value SetRbH [0] 0 = Do not suppress automatic mode and do not switch on de-icing mode manually 2 = Suppress automatic mode of blade heating system 8 = Switch on de-icing mode manually 10 = Suppress automatic mode and switch on de-icing mode manually 128 = Switch on blade heating system for pre-defined heating time The set value can also be retrieved for review. Private key SetRbH [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetRbH [2] The public key made available by the server after logging on to the session.										
RbH	Current status of the blade heating system (integer/read access). The status of the blade heating system is transmitted to SCADA as a 16-bit word. Each bit stands for a status. In order to be able to analyse the status indicated by the 'RbH' item, the decimal number needs to be converted to binary code. The following table indicates the statuses allocated to the bits: <table> <tr> <td>Bit 0 = 1</td><td>Automatic rotor blade de-icing allowed (parameter 1314)</td></tr> <tr> <td>Bit 1 = 1</td><td>Automatic BHS mode suppressed (SCADA)</td></tr> <tr> <td>Bit 2 = 1</td><td>BHS manually on (in wind energy converter)</td></tr> <tr> <td>Bit 3 = 1</td><td>BHS manually on (in SCADA)</td></tr> <tr> <td>Bit 4 = 1</td><td>Automatic rotor blade de-icing of the stopped WEC</td></tr> </table>	Bit 0 = 1	Automatic rotor blade de-icing allowed (parameter 1314)	Bit 1 = 1	Automatic BHS mode suppressed (SCADA)	Bit 2 = 1	BHS manually on (in wind energy converter)	Bit 3 = 1	BHS manually on (in SCADA)	Bit 4 = 1	Automatic rotor blade de-icing of the stopped WEC
Bit 0 = 1	Automatic rotor blade de-icing allowed (parameter 1314)										
Bit 1 = 1	Automatic BHS mode suppressed (SCADA)										
Bit 2 = 1	BHS manually on (in wind energy converter)										
Bit 3 = 1	BHS manually on (in SCADA)										
Bit 4 = 1	Automatic rotor blade de-icing of the stopped WEC										

Item	Description
	Bit 5 = 1 Automatic rotor blade de-icing during operation
	Bit 6 = 1 Blade heating system switched on as preventive measure (by automatic mode)
	Bit 7 = 1 Heating of rotor blades in a stopped WEC (SCADA)
	Bit 8 = 1 Heating of rotor blades while the WEC is running (SCADA)
	Bit 9 = 1 Free
	Bit 10 = 1 No power consumption available
	Bit 11 = 1 BHS fault
	Bit 12 = 1 Rotor blade de-icing allowed while the WEC is running (parameter 1315)
	Bit 13 = 1 Preventive heating of rotor blades allowed (parameter 1318)
	Bit 14 = 1 Free
	Bit 15 = 1 Blade heating system installed
	If the item value is 65636, no blade heating system is installed.
SetIceDet	Array for manually switching ice warning light on and off (array of 3 long words/read + write access). The array contains the following values in the specified sequence: Control value SetIceDet [0] 0 = Switch off ice warning light 8 = Switch on ice warning light The configured value can be viewed. Private key SetIceDet [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetIceDet [2] The public key made available by the server after logging on to the session.
IceDet	Current status of ice detection system (long word/read access). 0 = No ice detected 1 = Ice detection system using power curve method 2 = Preventive detection 4 = External sensor for ice detection system (Labko) 8 = External SCADA control signal (third party; external) 16 = Wind-farm ice detection system The status indicates which system/method detected the icing.

Tab. 74: Timings for access to individual wind energy converters for control interventions

Normal timeout during a session	60 s
Extended timeout after setting a parameter/ control value	60 s
Session-end timeout (delay for new reservation after stop) ¹	360 s
Session-end timeout (delay for new reservation after start) ¹	0 s
Error timeout (after three incorrect entries)	180 s
User ID error timeout	300 s

¹ The optional, asymmetrical session-end timeout can be activated by ENERCON Service.

4.1.2 Session items for wind farm start/stop of all wind energy converters in a wind farm

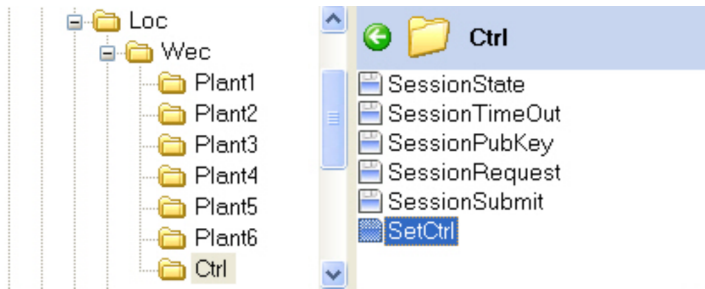


Fig. 35: Session items for the wind farm start/stop of all wind energy converters in a wind farm

The open-loop wind farm control of wind energy converters takes into account the current control status of each individual wind energy converter. A uniform status for all plants cannot be determined and therefore cannot be displayed. For safe operation, the control status of the relevant plant should be evaluated before performing any switching operation. Transmission takes place in a single session.

The open-loop wind farm control items reside in the *Loc\Wec\Ctrl* node.

Tab. 75: Session items for the wind farm start/stop of all wind energy converters in a wind farm

Item	Description
SetCtrl	Array for setting a control value (array of 3 long words/read + write access). The array contains the following values in the specified sequence: Control value SetCtrl [0] 0 = Start wind farm / put wind farm into operation 1 = Stop wind farm at 60° blade angle 2 = Stop wind farm at 90° blade angle 3 = Gradient stop at 60° blade angle 4 = Gradient stop at 90° blade angle 5 = Ice detection stop 6 = Shadow casting stop 7 = Animal protection stop at 60° blade angle 8 = Animal protection stop at 90° blade angle The control value can also be retrieved for review. Private Key SetCtrl [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetCtrl [2] The public key made available by the server after logging on to the session.

Tab. 76: Timings for the wind farm start/stop of all wind energy converters in a wind farm

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session-end timeout (delay for new reservation)	180 s
Error timeout (after three incorrect entries)	180 s
User ID error timeout	300 s

4.1.3 Session items for resetting the control system of individual wind energy converters

The *Loc\Wec\Plant<N>\Reset* node contains the items for resetting the control system of individual wind energy converters.

Transmission takes place in a single session.

Tab. 77: Session item for resetting the control system of individual wind energy converters
3.

Item	Description
 	Array for setting a control value. (Array of 3 long words/read + write access.) The array contains the following values in the specified sequence: Control value SetReset [0] Control value = plant number Private key SetReset [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetReset [2] The public key made available by the server after logging on to the session.

Tab. 78: Timings for resetting the control system of individual wind energy converters

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session-end timeout (delay for new reservation)	1800 s
Error timeout (after three incorrect entries)	180 s
User ID error timeout	300 s

4.2 Items for wind farm closed-loop and open-loop control (ENERCON SCADA Server)



The items provided in SCADA PDI-OPC are based on the SCADA control system; i.e., the item path and the items contained in it change according to the SCADA control settings. Interpretation of the items depends on the respective control system type.

4.2.1 Weekly table control

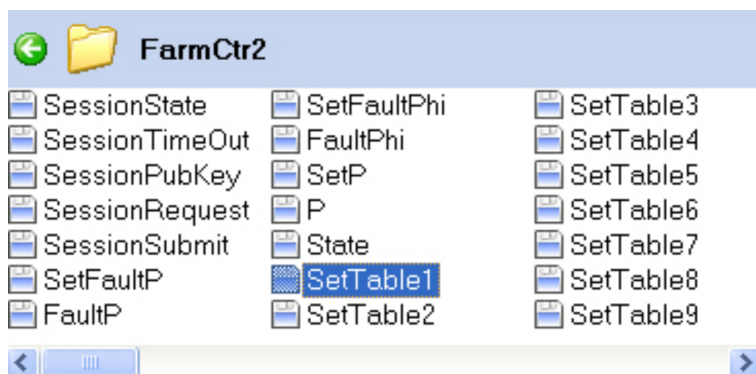


Fig. 36: Weekly table control items

Weekly table control (no closed-loop control) specifying the wind farm's relative power setpoint (0-100 %) and/or the phase angle of the injected power is performed in a single session. The relevant items reside in the *Scada\TableCtr1* node.

Tab. 79: Session items for weekly table control

Item	Description
TableX	Current table entry X of 40 for specifying power and phase angle setpoints (array of 8 doubles/read access). The array contains the following values in the sequence defined below: Day of week 1 = Sunday to 7 = Saturday Hour Hour = 0 to 23 Minute Minute = 0 to 59 Second Second = 0 to 59 Power setpoint Relative power setpoint in % = 0 to 100 Phase angle setpoint Absolute phase angle setpoint in °
SetTableX	Array for setting table entry X of 40 for specifying power and phase angle setpoints (array of 6 doubles/read access). The array contains the following values in the sequence defined below:

Item	Description
	Day of week 1 = Sunday to 7 = Saturday Hour Hour = 0 to 23 Minute Minute = 0 to 59 Second Second = 0 to 59 Power setpoint Relative power setpoint in % = 0 to 100 Phase angle setpoint Absolute phase angle setpoint in ° Private Key The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.
	Table entries are deleted by writing value day of week = hour = minute = second = 0.

Tab. 80: Timings for weekly table control

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session-end timeout (delay for new reservation)	150 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.2.2 Farm Control 2

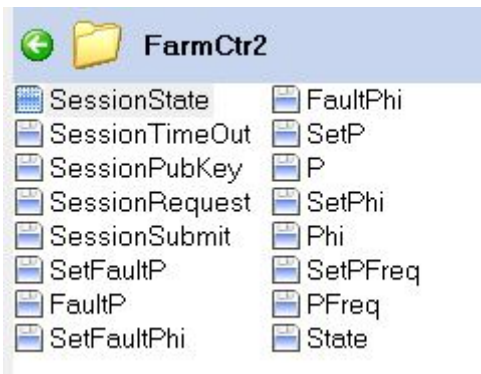


Fig. 37: Farm control 2 items

Closed-loop wind farm control data with the measuring point at the network connection point and setpoint specifications in the SCADA system are transmitted in a single session. Optionally, weekly table control can be activated for setpoint specification.

The absolute power setpoint of closed-loop control will then result from the relative table value and the absolute power setpoint.

$$P_{\text{control}} = (P_{\text{abs}} \times P_{\text{table}}) / 100$$

Transmitted in a single session. The relevant items reside in the *Scada\FarmCtr2* node.

Tab. 81: Closed-loop wind farm control 2 session items

Item	Description
State	Status of closed-loop control system. 0 = Normal operation.
FaultP	Current, relative default power value during a fault (e.g. communication fault) in the controller (double/read access)
SetFaultP	Array for setting the default power value (array of 3 doubles/read & write access). The array contains the following values in the specified sequence: Default power value SetFaultP [0] Relative default power value in %. The set value can also be retrieved for review. Private key SetFaultP [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetFaultP [2] The public key made available by the server after logging on to the session.
FaultPhi	Current, absolute default phase-angle value during a fault (e.g. communication fault) in the controller (array of 3 doubles/read + write access)

Item	Description
SetFaultPhi	Array for setting the default phase-angle value (array of 3 doubles/read + write access). The array contains the following values in the specified sequence: Default phase-angle value SetFaultPhi [0] Absolute default phase-angle value in °. The set value can also be retrieved for review. Private key SetFaultPhi [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetFaultPhi [2] The public key made available by the server after logging on to the session.
P	Absolute power setpoint (double/read access)
SetP	Array for setting the power setpoint (array of 3 doubles/read + write access). The array contains the following values in the specified sequence: Power setpoint SetP [0] Absolute power setpoint in kW. The set value can also be retrieved for review. Private key SetP [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetP [2] The public key made available by the server after logging on to the session.
Phi	Current, absolute phase-angle setpoint in ° (double/read) This item will not exist if a weekly table control is active.
SetPhi	Array for setting the phase-angle setpoint in ° (array of 3 doubles/read + write access). This item will not exist if a weekly table control is active. The array contains the following values in the specified sequence: Phase-angle setpoint SetPhi [0] Phase-angle setpoint in °

Item	Description
	Private key SetPhi [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetPhi [2] The public key made available by the server after logging on to the session.
SetPFreq	Array for setting the frequency reserve capacity (array of 3 doubles/read + write access). The array contains the following values in the specified sequence: Power setpoint SetPFreq[0] Relative control value for frequency reserve capacity in %. The set value can also be retrieved for review. Private key SetPFreq[1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetPFreq[2] The public key made available by the server after logging on to the session.
PFreq	Current, relative frequency-reserve capacity setpoint (double/read access)

Tab. 82: Timings for closed-loop wind farm control 2

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session-end timeout (delay for new reservation)	150 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.2.3 Apparent power control 2

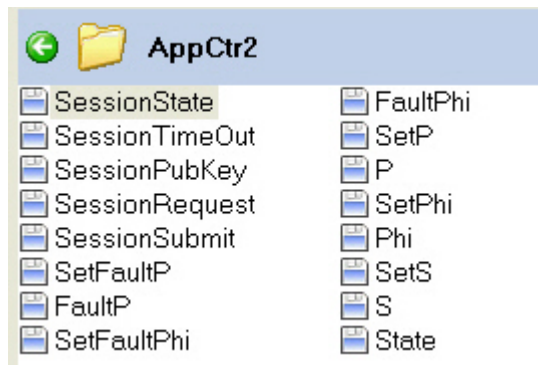


Fig. 38: Items for apparent-power control 2

Closed-loop wind farm control data with the measuring point at the network connection point and setpoint specifications in the SCADA system are transmitted in a single session. Optionally, weekly table control can be activated as a setpoint specification.

The absolute power setpoint of the closed-loop control system is then derived from the relative table value and the absolute power setpoint:

$$P_{\text{control}} = (P_{\text{abs}} \times P_{\text{table}})/100$$

The apparent-power setpoint is an apparent-power limit. Reactive power (Phi) will be reduced once this limit has been reached.

All data points of closed-loop wind farm control 2 are also present for apparent power control 2. Only the apparent-power limit is added.

The relevant items reside in the *ScadaAppctr2* node.

Tab. 83: Session items of apparent power control 2

Item	Description
S	Absolute apparent power setpoint limit (double/read access)
SetS	Array for setting the apparent power limit (array of 3 doubles/read access + write access). The array contains the following values in the specified sequence: Apparent power limit SetS [0] Absolute apparent power limit in kVA. The set value can also be retrieved for review. Private key SetS [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetS [2] The public key made available by the server after logging on to the session.

Tab. 84: Timings of apparent-power control 2

Normal timeout during a session	60 s
Extended timeout after setting a parameter/ control value	60 s
Session-end timeout (delay for new reservation)	150 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.2.4 Open Loop Control (OIctr1)

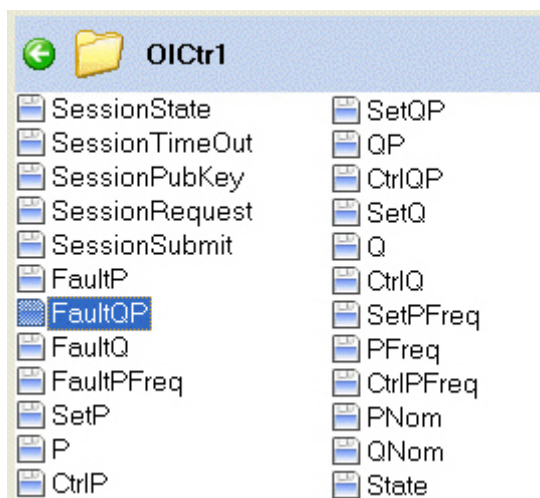


Fig. 39: Open-loop control (OIctr1) items

Open-loop control provides open-loop, not closed-loop, control of the wind farm. The set-points specified via OPC are checked and forwarded with limitations to the wind energy converter in a fixed cycle. No additional hardware (RTU or FCU) is required for open-loop control via OPC. With this type of control, the OPC, and thus also the client connection, is integrated into the controller cycle and monitored.

The integration of the client connection into the controller cycle presupposes a high availability of the connection between client and server. The second network interface of the ENERCON SCADA Server can be used for this. The availability of this network interface needs to be checked for each specific project.

Data is transmitted in a loop session. The relevant items reside in the *Scada\OIctr1* node. Information on the loop session can be obtained from the document D0137978 "Technische Beschreibung ENERCON SCADA PDI-OPC" (Technical description of ENERCON SCADA PDI-OPC).

Tab. 85: Open-loop control (OIctr1) session items

Item	Description
FaultP	Current, relative default value during a fault in the controller (double/read access)
FaultQP	Current, absolute Q/P default value during a fault in the control system (double/read access)

Item	Description
FaultQ	Current, relative Q default value during a fault in the control system (double/read access)
FaultPFreq	Current, relative default value for frequency reserve capacity during a fault in the control system (double/read access)
SetP	Array for setting the power setpoint (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Power setpoint SetP [0] Relative power setpoint in %. The set value can also be retrieved for review. Private key SetP [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetP [2] The public key made available by the server after logging on to the session.
P	Current, relative power setpoint (double/read access)
CtrlP	Current, relative P control value (double/read access)
SetQP	Array for setting the Q/P setpoint (array of 3 doubles/read & write access). This array contains the following values in the sequence defined below: Power setpoint SetQP [0] Q/P setpoint The set value can also be retrieved for review. Private key SetQP [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetQP [2] The public key made available by the server after logging on to the session.
QP	Current, absolute Q/P setpoint (double/read access)
CtrlQP	Current Q/P control value (double/read access)
SetQ	Array for setting the reactive power setpoint (array of 3 doubles/read + write access). This array contains the following values in the sequence defined below:

Item	Description
	Power setpoint SetQ [0] Relative reactive power setpoint in %. The set value can also be retrieved for review. Private key SetQ [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetQ [2] The public key made available by the server after logging on to the session.
Q	Current, relative reactive power setpoint (double/read access)
CtrlQ	Current, relative Q control value (double/read access)
SetPFreq	Array for setting the frequency reserve capacity (array of 3 doubles/read + write access). The array contains the following values in the specified sequence: Power setpoint SetPFreq [0] Relative control value for frequency reserve capacity in %. The set value can also be retrieved for review. Private key SetPFreq [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetPFreq [2] The public key made available by the server after logging on to the session.
PFreq	Current, relative frequency-reserve capacity setpoint (double/read access)
CtrlPFreq	Current control value for frequency reserve capacity (double/read access)
PNom	Nominal power of the wind farm (double/read access)
QNom	Nominal reactive power of the wind farm (double/read access)
State	Status of closed-loop control system 0 = Normal operation 144 = Controller fault 246 = Communication with client disrupted

Tab. 86: Open-loop control (OICtr1) timings

Normal timeout during a session	10 x control cycle
Extended timeout after setting a parameter/ control value	10 x control cycle
Session-end timeout (delay for new reservation)	60 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.2.5 Manual control 1 (ManCtr1)

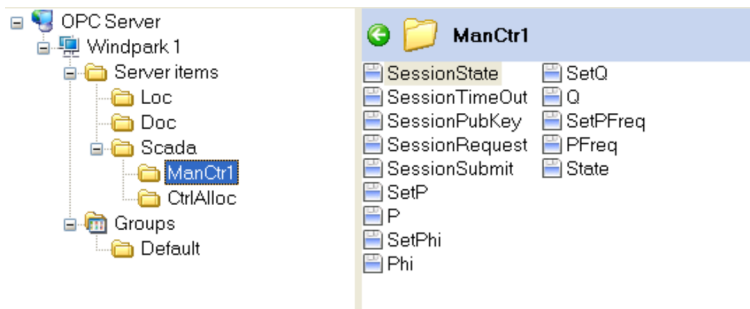


Fig. 40: Items for manual control 1

Tab. 87: Session items for manual control 1 (ManCtr1)

Item	Description
SetP	Array for setting the power setpoint (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Power setpoint SetP [0] Relative power setpoint in %. The set value can also be retrieved for review. Private key SetP [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetP [2] The public key made available by the server after logging on to the session.
P	Current, relative power setpoint (double/read access)
SetPhi	Array for setting the phase-angle setpoint (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Phase-angle setpoint SetPhi [0] Phase-angle setpoint in °. The set value can also be retrieved for review. Private key SetPhi [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetPhi [2]

Item	Description
	The public key made available by the server after logging on to the session.
Phi	Current phase-angle setpoint (double/read access)
SetQ	Array for setting the reactive power setpoint (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Reactive power setpoint SetQ [0] Relative reactive power setpoint in %. The set value can also be retrieved for review. Private key SetQ [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetQ [2] The public key made available by the server after logging on to the session.
Q	Current, relative reactive power setpoint (double/read access)
SetPFreq	Array for setting the setpoint for the frequency reserve capacity (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Power setpoint SetPFreq [0] Setpoint for relative frequency reserve capacity in %. The set value can also be retrieved for review. Private key SetPFreq [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetPFreq [2] The public key made available by the server after logging on to the session.
PFreq	Current, relative power setpoint (double/read access)
State	Status of the open-loop control system. 0 = Normal operation

Tab. 88: Timings for manual control 1 (ManCtr1)

Normal timeout during a session	60 s
Extended timeout after setting a parameter/ control value	60 s
Session-end timeout (delay for new reservation)	150 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.2.6 Power Consumption Management

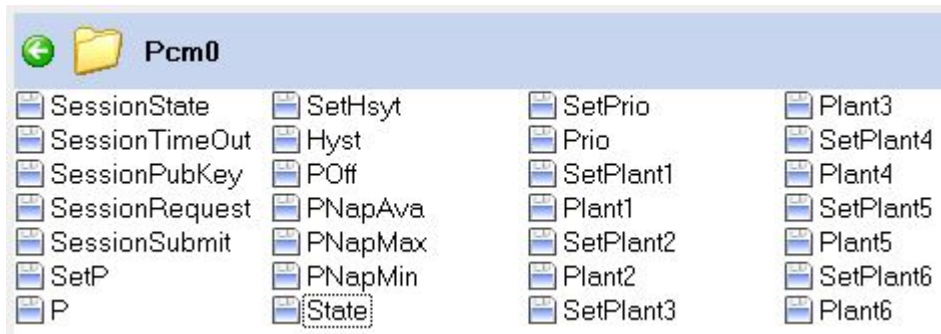


Fig. 41: Items for power consumption management (six WECs in the group)

The Power Consumption Management data is transmitted in a single session. The relevant items reside in the *SCADA\Pcm<N>* bus node.

A bus node with the relevant items is created for each active plant group. The number of *SetPlant* and *Plant* items corresponds to the number of wind energy converters in the group.

The following screen shot shows an example of these items for a group with six wind energy converters.

Tab. 89: Session items for the power consumption management

Item	Description
SetP	Array for setting the power-consumption setpoint (array of 3 long words/read + write access). The array contains the following values in the sequence defined below: Power-consumption setpoint SetP [0] Power-consumption setpoint in kW. The set value can also be retrieved for review. Private key SetP [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetP [2] The public key made available by the server after logging on to the session.
P	Current power-consumption setpoint (integer/read access)
SetHyst	Array for setting the hysteresis (array of 3 long words/read + write access). The array contains the following values in the sequence defined below: Hysteresis setpoint SetHyst [0] Hysteresis setpoint in %. The set value can also be retrieved for review.

Item	Description
	Private key SetHyst [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetHyst [2] The public key made available by the server after logging on to the session.
Hyst	Current hysteresis (integer/read access)
POff	Currently available power (integer/read access)
PNapAva	Mean power consumption measured at the NCP or determined by a virtual NCP (integer/read access)
PNapMax	Maximum power consumption measured at the NCP or determined by a virtual NCP (integer/read access).
PNapMin	Maximum power consumption measured at the NCP or determined by a virtual NCP (integer/read access).
State	Status of the power consumption management 0 = Start consume program 1 = End consume program 60 = Parameters changed 61 = WEC priority changed 62 = WEC mode changed 105 = Data-logger sensor error 215 = WEC error, measurement value 246 = NCP error, sensor measurement value
SetPrio	Array for setting the priority (array with variable size of n+2 long words/read + write access; n = number of wind energy converters in the group). The array contains the following values in the sequence defined below: Places in the list (1 to n) SetPrio [0] to SetPrio [n-1] Plant number of the wind energy converters. The set values can also be retrieved for review. Private key SetPrio [n] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetPrio [n+1]

Item	Description
	The public key made available by the server after logging on to the session.
Prio	Currently set priority and/or call sequence (array with variable size n word/read access; n = number of wind energy converters in the group). Prio [0] = Wind energy converter 1; Prio [1] = Wind energy converter 2, etc.
SetPlant<N>	Array for selecting the operating mode of the wind energy converter (array of 4 long words/read + write access). The array contains the following values in the sequence defined below: Operating mode SetPlant<N> [0] Operating mode of the wind energy converter's power consumption management. 0 = Automatic mode 1–253 = Manual mode; value corresponds to duration in hours. The set value can also be retrieved for review. Power-consumption availability SetPlant<N> [1] Power offered for consumption for the wind energy converter's manual operating mode. The set value can also be retrieved for review. Private key SetPlant<N> [2] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetPlant<N> [3] The public key made available by the server after logging on to the session.
Plant<N>	Array for reading current information from the wind energy converter (array of 8 long words/read access). The array contains the following values in the sequence defined below: Maintenance Plant<N> [0] 1 = Wind energy converter under maintenance Error code Plant<N> [1] Indicates any possible error as a numerical code. Operating mode Plant<N> [2] Current operating mode of the power consumption management in the respective wind energy converter.

Item	Description
	Deactivated = Wind energy converter is logged off from power consumption management (e.g. for maintenance purposes).
	Sync = Synchronisation, e.g. after a restart of the wind energy converter
	Error = Communication problem (read-only)
	Time
	Plant<N> [3]
	Remaining duration of manual mode
	Power-consumption availability
	Plant<N> [4]
	Power currently offered for consumption
	Reserved
	Plant<N> [5]
	Power currently reserved by the wind energy converter
	Reserved (t-1)
	Plant<N> [6]
	Power reserved for consumption in previous cycle
	Desired power
	Plant<N> [7]
	Power required by the wind energy converter

Tab. 90: Timings for power consumption management

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session-end timeout (delay for new reservation)	300 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.3 Items for closed-loop and open-loop RTU/FCU/FCU E2 control systems

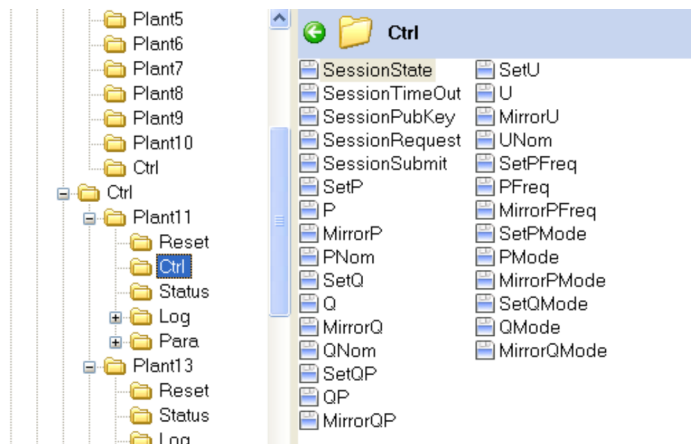


Fig. 42: The *Loc\Ctrl\Plant<N>\Ctrl* node for RTU, FCU or FCU E2

Using the items described below, setpoints can be transmitted to the ENERCON SCADA system's wind-farm control unit (RTU, FCU, FCU E2). In addition, the current setpoints can be read out.

Individual control functions are activated/deactivated directly on the ENERCON SCADA system's farm control unit.

The *Loc\Ctrl\Plant<N>\Ctrl* node contains items for RTU, FCU or FCU E2 control access.



Alternatively, the power factor (Cos Phi), the phase angle Phi, or Q/P can be transmitted as a setpoint for reactive-power control. This setting has to be made on the OPC server.

Tab. 91: Session items for RTU, FCU or FCU E2

Item	Description
SetP	Array for setting the active-power setpoint (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Power setpoint SetP [0] Relative active-power setpoint in %. Minimum value: 0 % Maximum value: 100 % The set value can also be retrieved for review. Private key SetP [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetP [2] The public key made available by the server after logging on to the session.

Item	Description
P	Relative active-power setpoint P1 – return (RTU/FCU/FCU E2) [% of P_{Nom} for wind farm] (double/read access)
MirrorP	Active-power setpoint – OPC return (double/read access)
PNom	Nominal power of the wind farm in kW (RTU/FCU) (double/read access) Grid connection capacity of the wind farm in kW (FCU E2) (double/read access)
SetQ	Array for setting the reactive-power setpoint (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Reactive-power setpoint SetQ [0] Relative reactive-power setpoint in %. Minimum value: -100 % Maximum value: +100 % The set value can also be retrieved for review. Private key SetQ [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetQ [2] The public key made available by the server after logging on to the session.
Q	Reactive-power setpoint – return (RTU/FCU/FCU E2) [% of P_{Nom} for wind farm] (double/read access)
MirrorQ	Reactive-power setpoint – OPC return (double/read access)
QNom	Nominal reactive power of the wind farm in kVAr (double/read access)
SetCos(Phi)* Alternatively: SetQ/P, SetPhi	Array for setting the power-factor setpoint (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Power-factor setpoint SetCos(Phi) [0] Minimum value: -1 Maximum value: +1 The set value can also be retrieved for review. Private key SetCos(Phi) [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients.

Item	Description
	Public key SetCos(Phi) [2] The public key made available by the server after logging on to the session. Phase-angle setpoint Phi Minimum value: -90° Maximum value: +90° The structure and name of the respective array correspond to the other setpoints. Power-factor setpoint Q/P Minimum value: -3 Maximum value: +3 The structure and name of the respective array correspond to the other setpoints.
Cos(Phi) Alternatively: Q/ P, Phi	Power-factor setpoint – return (RTU/FCU/FCU E2) [% of P_{Nom} for wind farm] (double/read access)
MirrorCos(Phi) Alternatively: Mir- rorQP, MirrorPhi	Power-factor setpoint – OPC return (double/read access)
SetU	Array for setting the voltage setpoint (array of 3 doubles/read + write access). The array contains the following values in the sequence defined below: Voltage setpoint SetU [0] Relative voltage setpoint in %. Minimum value: 0 % Maximum value: 200 % A value of 100 % corresponds to the nominal voltage. The set value can also be retrieved for review. Private key SetU [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetU [2] The public key made available by the server after logging on to the session.
U	Voltage setpoint – return (RTU/FCU/FCU E2) [% of P_{Nom} for wind farm] (double/read access)

Item	Description
MirrorU	Voltage setpoint – OPC return (double/read access)
UNom	Nominal voltage of the wind farm in V (double/read access)
SetPFreq	Array for setting the frequency reserve capacity (array of 3 doubles/read + write access). The array contains the following values in the specified sequence: Power setpoint SetPFreq [0] Relative control value for frequency reserve capacity in %. Minimum value: 50 % Maximum value: 100 % If the value is 100 %, no reserve capacity is made available. The set value can also be retrieved for review. Private key SetPFreq [1] The private key used by the client to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public key SetPFreq [2] The public key made available by the server after logging on to the session.
PFreq	Frequency reserve-capacity setpoint – return (RTU/FCU/FCU E2) (double/read access)
MirrorPFreq	Frequency reserve-capacity setpoint – OPC return (double/read access)
SetPMode	Setpoint for Active Power Control mode
PMode	Setpoint for Active Power Control mode – return (RTU/FCU/FCU E2)
MirrorPMode	Setpoint for Active Power Control mode – return (OPC)
SetQMode	Setpoint for Reactive Power Control mode
QMode	Setpoint for Reactive Power Control mode – return (RTU/FCU/FCU E2)
MirrorQMode	Setpoint for Reactive Power Control mode – return (OPC)

Tab. 92: Timings of the RTU/FCU/FCU E2 closed-loop and open-loop control systems

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session-end timeout (delay for new reservation)	60 s ¹
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

¹ For direct marketing, the session-end timeout can be set to 10 s.

List of tables

Tab. 1	Standard data.....	7
Tab. 2	15-minute data (Wec15, Rtu15, Farmctr15) (optional).....	7
Tab. 3	Availability data (Ava).....	7
Tab. 4	Temperature data (TXX)	7
Tab. 5	Loc node	9
Tab. 6	Loc\Wec node	10
Tab. 7	The Loc\Wec\Plant<N> node	12
Tab. 8	Node Loc\Wec\Plant<N>\Status – Items.....	14
Tab. 9	Node Loc\Wec\Plant<N>\Status – Elements of the array	15
Tab. 10	The Loc\Wec\Plant<N>\Log node	16
Tab. 11	Loc\Wec\Plant<N>\Log\Wec15 node (optional)	17
Tab. 12	Loc\Wec\Plant<N>\Log\Wec15\<Period> node (optional)	18
Tab. 13	Loc\Wec\Plant<N>\Log\T<XX> node	19
Tab. 14	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS40 (T40).....	20
Tab. 15	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS44a (T44a).....	20
Tab. 16	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS44b (T44b).....	21
Tab. 17	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS48 (T48a).....	21
Tab. 18	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS66 (T66).....	22
Tab. 19	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS70a (T70a).....	23
Tab. 20	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS70b (T70b).....	24
Tab. 21	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS82 (T82a1).....	24
Tab. 22	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS82 (T82a2).....	25
Tab. 23	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS101 (T101a1).....	25
Tab. 24	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS101 (T101a2).....	26
Tab. 25	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS126 (T126a1).....	27
Tab. 26	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS126 (T126a2).....	28
Tab. 27	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for CS126 (T126a3).....	28
Tab. 28	The Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for EP3-CS-02 (Tep3c021)	29
Tab. 29	The Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for EP3-CS-02 (Tep3c022)	30
Tab. 30	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for EP4-CS-01 (Tep4c011).....	30
Tab. 31	Loc\Wec\Plant<N>\Log\T<XX>\<Period> node for EP4-CS-01 (Tep4c012).....	31
Tab. 32	Loc\Wec\Plant<N>\Log\Wecstd node	33
Tab. 33	Loc\Wec\Plant<N>\Log\Wecstd\<Period> node.....	34
Tab. 34	The Loc\Wec\Plant<N>\Log\Weceolog node (optional)	36
Tab. 35	The Loc\Wec\Plant<N>\Log\Weceolog\<Period> node (optional)	36
Tab. 36	The Loc\Wec\Plant<N>\Log\Wecfos4x node (optional)	39
Tab. 37	The Loc\Wec\Plant<N>\Log\Wecfos4x\<Period> node (optional).....	39
Tab. 38	The Loc\Wec\Plant<N>\Log\Wecwoef node (optional)	40
Tab. 39	The Loc\Wec\Plant<N>\Log\Wecwoef\<Period> node (optional).....	40
Tab. 40	Node Loc\Wec\Plant<N>\Ava – Nodes and items	41

Tab. 41	Node <i>Loc\Wec\Plant<N>\Ava</i> – Elements of the array	41
Tab. 42	<i>Loc\Wec\Plant<N>\Ava\<Period></i> node	43
Tab. 43	<i>Loc\Sens\Plant<N></i> node	44
Tab. 44	<i>Loc\Sens\Plant<N>\Log\Mspgerf1\<Period></i> node.....	45
Tab. 45	<i>Loc\Conv\Plant<N></i> node of an ENERCON solar converter	46
Tab. 46	The <i>Loc\Conv\Plant<N>\Log\Solar1\<Period></i> node.....	47
Tab. 47	<i>Loc\Conv\Plant<N></i> node of the ENERCON feeding and charging unit.....	49
Tab. 48	<i>Loc\Conv\Plant<N>\Log\Bcs1\<Period></i> node	50
Tab. 49	<i>Loc\Conv\Plant<N>\Log\Bcs2\<Period></i> node	52
Tab. 50	<i>Loc\Ctrl\Plant<N></i> node of the RTU	56
Tab. 51	<i>Loc\Ctrl\Plant<N>\Log\Rtu1\<Period></i> node.....	58
Tab. 52	<i>Loc\Ctrl\Plant<N>\Log\Rtu15\<Period></i> node (optional).....	60
Tab. 53	<i>Loc\Ctrl\Plant<N>\Log\Rtuphase1\<Period></i> node.....	61
Tab. 54	<i>Loc\Ctrl\Plant<N></i> node of the FCU	63
Tab. 55	<i>Loc\Ctrl\Plant<N>\Log\Fcuctr1\<Period></i> node	64
Tab. 56	<i>Loc\Ctrl\Plant<N></i> node of the FCU E2	66
Tab. 57	<i>Loc\Ctrl\Plant<N>\Log\Fcu2ctr1\<Period></i> node	67
Tab. 58	<i>Loc/Ctrl/Plant<N></i> node of the GDA	71
Tab. 59	<i>Loc\Ctrl\Plant<N>\Log\Farmctr1\Day</i> node	73
Tab. 60	<i>Loc\Ctrl\Plant<N>\log\Farmctr15<Period></i> node (optional).....	75
Tab. 61	<i>Loc\Ctrl\Plant<N></i> node of the SCU	76
Tab. 62	<i>Loc\Ctrl\Plant<N>\Log\Substat1\<Period></i> node	77
Tab. 63	<i>Loc/Ctrl/Plant<N></i> node for transmission substation 2	78
Tab. 64	<i>Loc\Ctrl\Plant<N>\Log\Substat2\<Period></i> node	80
Tab. 65	<i>Loc\Ctrl\Plant<N>\Log\SubTemp2\<Period></i> node.....	82
Tab. 66	<i>Loc\Ctrl\Plant<N>\Log\Subphase2\<Period></i> node	83
Tab. 67	<i>Loc\Meteo\Plant<N></i> node	84
Tab. 68	<i>Loc\Meteo\Plant<N>\Log\Wind1\<Period></i> node.....	85
Tab. 69	<i>Loc\Meteo\Plant<Index>\Log\Weather2\<Period></i> node	86
Tab. 70	<i>Loc/Actor/Plant<N></i> node of the PDI	87
Tab. 71	The <i>Scada</i> node	89
Tab. 72	Session items for control interventions	90
Tab. 73	Session item for access to individual wind energy converters for control interventions.....	92
Tab. 74	Timings for access to individual wind energy converters for control interventions.....	95
Tab. 75	Session items for the wind farm start/stop of all wind energy converters in a wind farm.....	96
Tab. 76	Timings for the wind farm start/stop of all wind energy converters in a wind farm.....	97
Tab. 77	Session item for resetting the control system of individual wind energy converters ...	97
Tab. 78	Timings for resetting the control system of individual wind energy converters	97
Tab. 79	Session items for weekly table control	98
Tab. 80	Timings for weekly table control	99

Tab. 81	Closed-loop wind farm control 2 session items	100
Tab. 82	Timings for closed-loop wind farm control 2.....	102
Tab. 83	Session items of apparent power control 2	103
Tab. 84	Timings of apparent-power control 2.....	104
Tab. 85	Open-loop control (OICtr1) session items	104
Tab. 86	Open-loop control (OICtr1) timings	107
Tab. 87	Session items for manual control 1 (ManCtr1).....	108
Tab. 88	Timings for manual control 1 (ManCtr1).....	110
Tab. 89	Session items for the power consumption management	111
Tab. 90	Timings for power consumption management	114
Tab. 91	Session items for RTU, FCU or FCU E2.....	115
Tab. 92	Timings of the RTU/FCU/FCU E2 closed-loop and open-loop control systems.....	118

Glossary of terms

Array	In computer science, an array is a data structure. Using a field, data of a typically standardised type can be stored in computer memory in an arranged manner. Data access is made possible by using an index.
ENERCON SCADA FCU	The superordinate wind farm control unit registers the voltage and the power injected at a reference point and uses these values to compute the actual values of the controlled variables, e.g. active and reactive power. The difference to the prescribed setpoints is used to generate the respective actuating variables, and they are sent to the wind turbines in the wind farm via a data bus.
ENERCON SCADA METEO	The ENERCON SCADA METEO system serves to acquire and analyse meteorological data in combination with ENERCON SCADA.
ENERCON SCADA PDI	Hardware-based interface for transmitting process data between different communication devices such as the ENERCON SCADA system and the grid operator. The grid operator can call up data for wind turbines and systems via ENERCON SCADA PDI. ENERCON provides different Process Data Interfaces.
ENERCON SCADA RTU-C	Wind farm controller that can control a wind farm relative to the actual parameter at the NCP.
ENERCON SCADA System	System for data acquisition, remote monitoring, and open-loop/closed-loop control both in individual wind energy converters and entire wind farms. The ENERCON SCADA system enables customers and ENERCON Service to monitor operating states and to analyse stored operating data. Furthermore, authorised users may use it to modify the operating parameters of the wind energy converters and the connection to the power grid. Depending on the application, the ENERCON SCADA system also provides additional options that enable closed-loop control based on setpoints (e.g. active power).
Flag	In computer science, a flag is a binary variable or a status indicator. A flag can be used to identify specific states.
GDA	The Grid Data Acquisition system (GDA) is used for measuring key electrical variables at the wind farm's point of common coupling. These values are transmitted to the ENERCON SCADA Server for display and control purposes as well as data storage.
Item	Within the context of OPC, the combination of a (physical) value and its properties is called an Item.
OPC XML-DA	Interface standard of the OPC Foundation that is used for XML-based exchange of data between computer systems, particularly via the Internet, among other things.
SCU	The Substation Control Unit (SCU) is a communication interface between the control system of the transmission substation and the ENERCON SCADA system.
Server	A server is a program that communicates with other programs and can provide special services and data to the latter.